

Beginner's Introduction to Computing at CARC with JupyterHub

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Example ESACPE 2023

Version 0.1

Goals

- 1) Basic Linux Literacy
- 2) Using Compute Nodes with Slurm
- 3) Jupyterhub

I hope to spend about an hour on each with a 15 minute break.

Outline

- High Performance Computing Overview
- Logging in
- The BASH Shell
- The Slurm Job Scheduler
- Transferring data to and from CARC
- Accessing software and the module system
- Conda environments
- Jupyterhub

High Performance Computing

- What is high performance computing?
 - Really just means something that is a lot more powerful than your desktop or laptop.
 - Hardware:
 - That might mean more and faster processors to do the calculations more quickly (eg 400 CPUs instead of 4)
 - More RAM so you can work on bigger problems (3,000 GB instead of 8)
 - Bigger file systems so you can process larger datasets
 - More and bigger GPUs to accelerate your computations (12 GPUs at a time instead of 1)
 - People:
 - Someone else to manage the systems and keep them running and secure
 - Someone to answer your questions and help with problems

Logging into Wheeler



First login to the Linux **workstation** in front of you.

Use your CARC username and password.

Keven, Tannor, Viacheslav (Slava), and Jose can help you login if you have trouble.

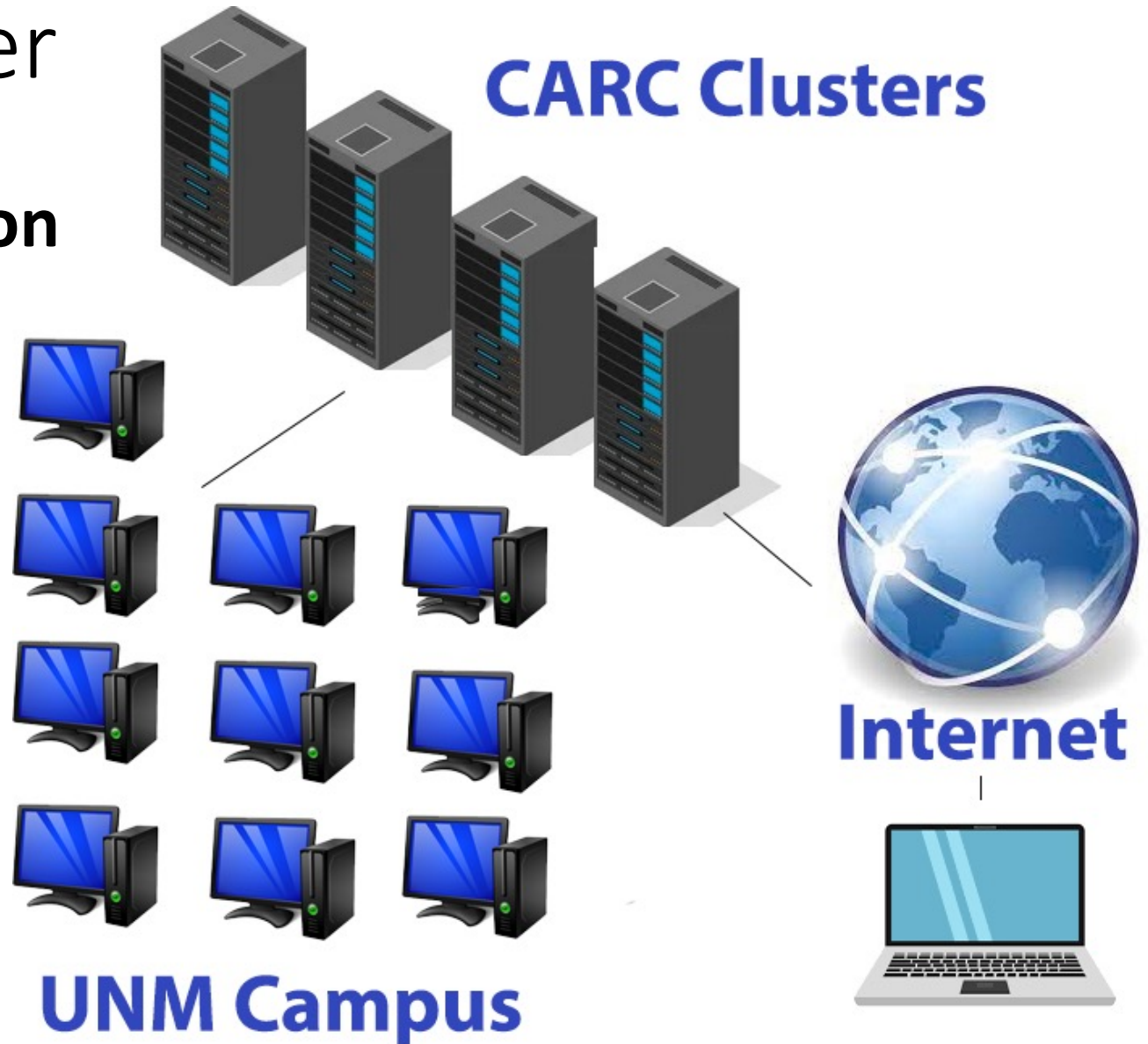
This is an “important step” so don’t let me move on until you have logged in

Logging into Wheeler

First login to the **workstation** in front of you.

You will always login to CARC cluster remotely.

These clusters don't even have monitors.



Logging into Wheeler

We are going to use a program called [secure shell](#).

[Secure shell \(ssh\)](#) is now built into every major operating system (Windows, OSX, and Linux).

You don't need third party programs like putty anymore.

Logging into Wheeler



```
ssh vanilla@wheeler.alliance.unm.edu
```

Should prompt you for a password...

Don't let me move on until you are able to login.

Logging into Wheeler

```
Matthew — ssh wheeler — 82x27
last login: Tue Jun 14 14:47:24 2022 from fricke.co.uk
-----
Welcome to Wheeler

Be sure to review the "Acceptable Use" guidelines posted on the CARC website.

For assistance using this system email help@carc.unm.edu.

Tutorial videos can be accessed through the CARC website: Go to
http://carc.unm.edu, select the "New Users" menu and then click
"Introduction to Computing at CARC".

Warning: By default home directories are world readable. Use the chmod command
to restrict access.

Don't forget to acknowledge CARC in publications, dissertations, theses and
presentations that use CARC computational resources:

"We would like to thank the UNM Center for Advanced Research Computing,
supported in part by the National Science Foundation, for providing the
research computing resources used in this work."

Please send citations to publications@carc.unm.edu.
-----
Starting SSH Key Agent...
agent pid 19486
fricke@wheeler:~ $
```

Please enter the following command



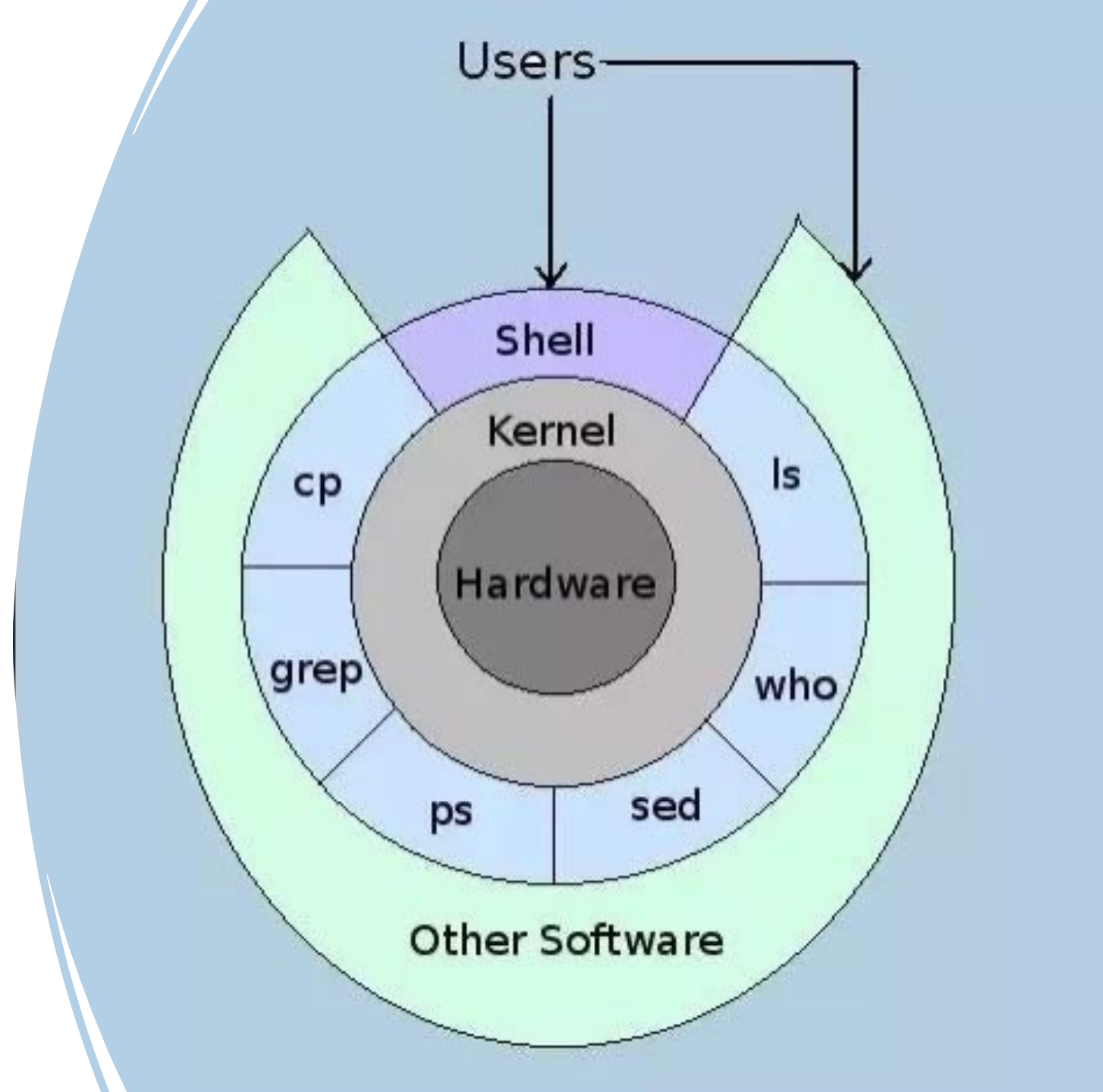
```
cp -r /projects/shared/workshops/beginner/mystuff ~/
```

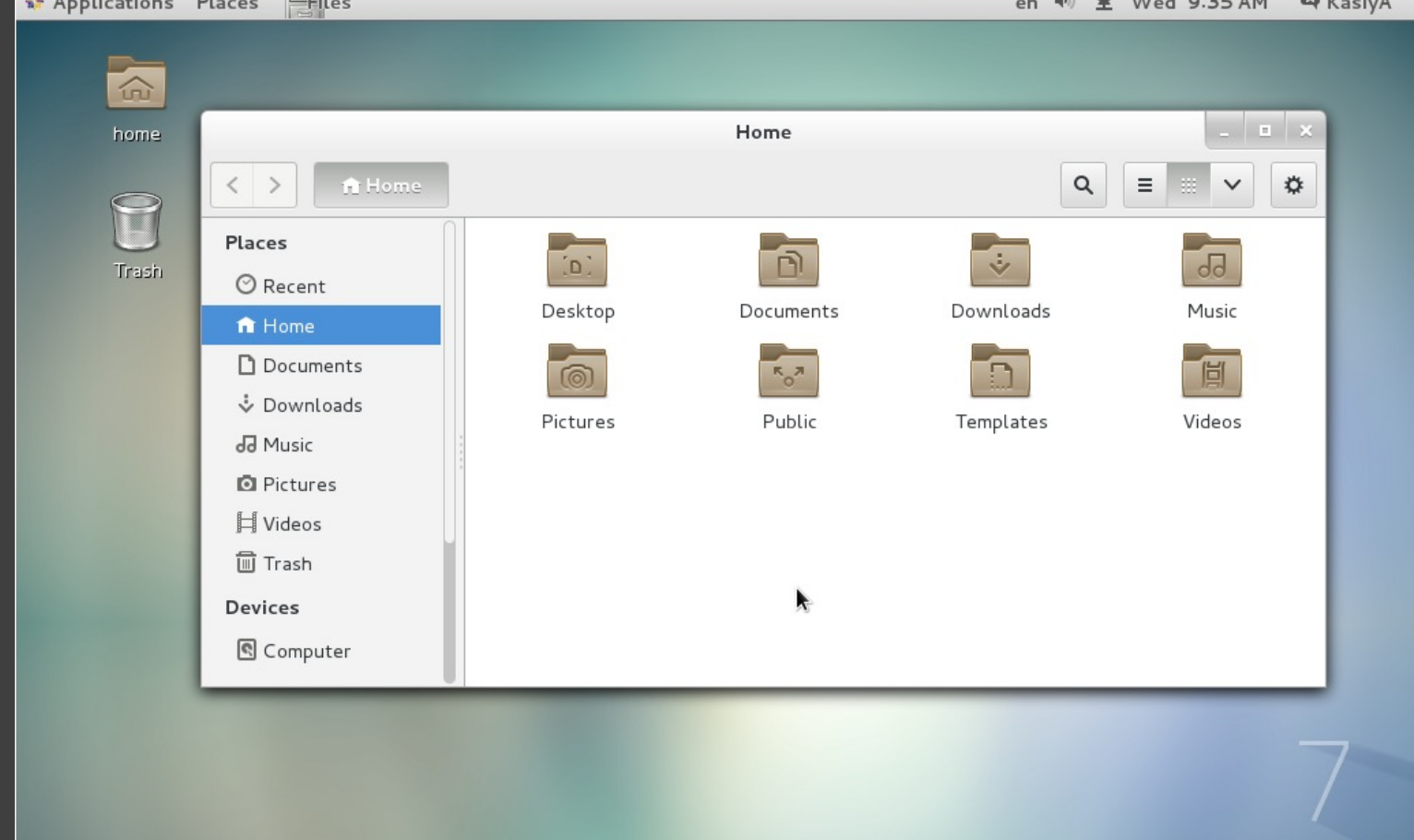
We will come help you if you have any trouble.

(Later I will go over what this command does)

Linux and the BASH Shell

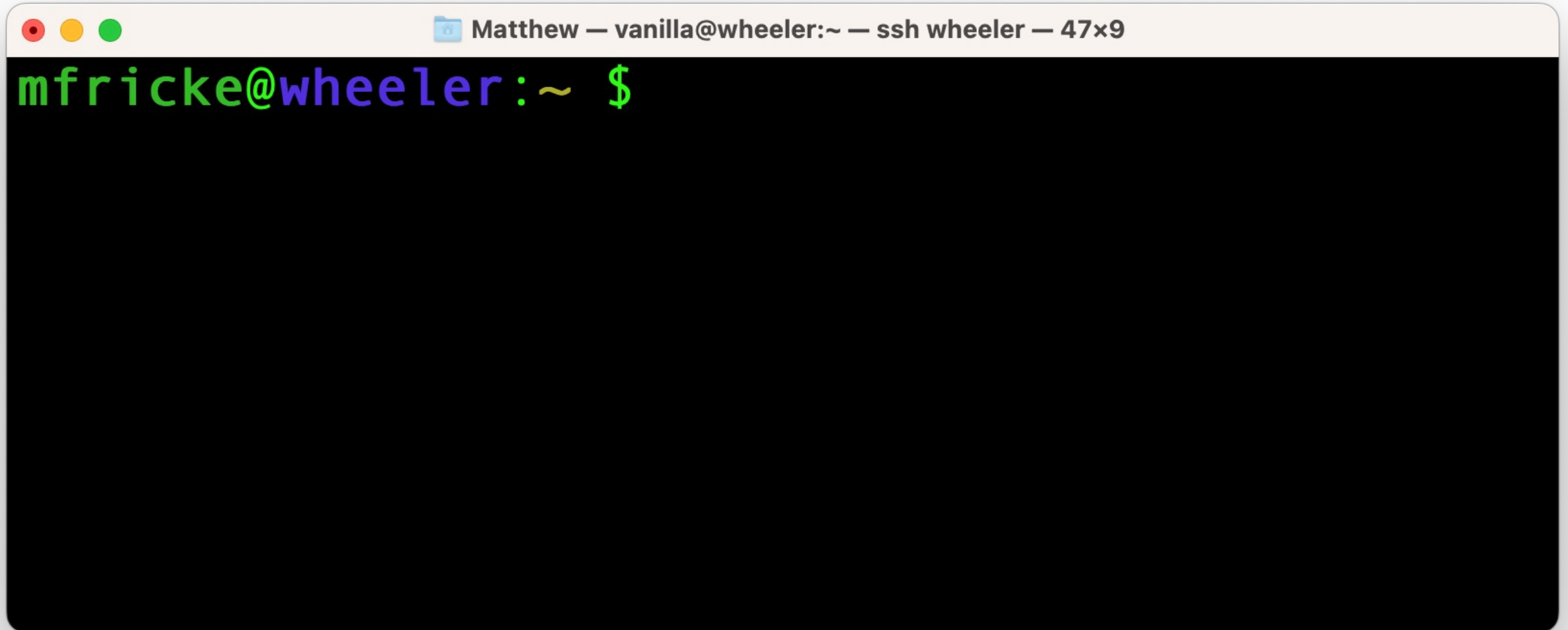
- The Kernel manages access to the hardware in a computer.
- An Operating System (OS) is the Kernel plus useful programs provided by the OS.
- The “shell” is the outermost layer of the OS.
- It is where the user interacts with the OS.





Graphical Shells (GUIs)

Logging into Wheeler



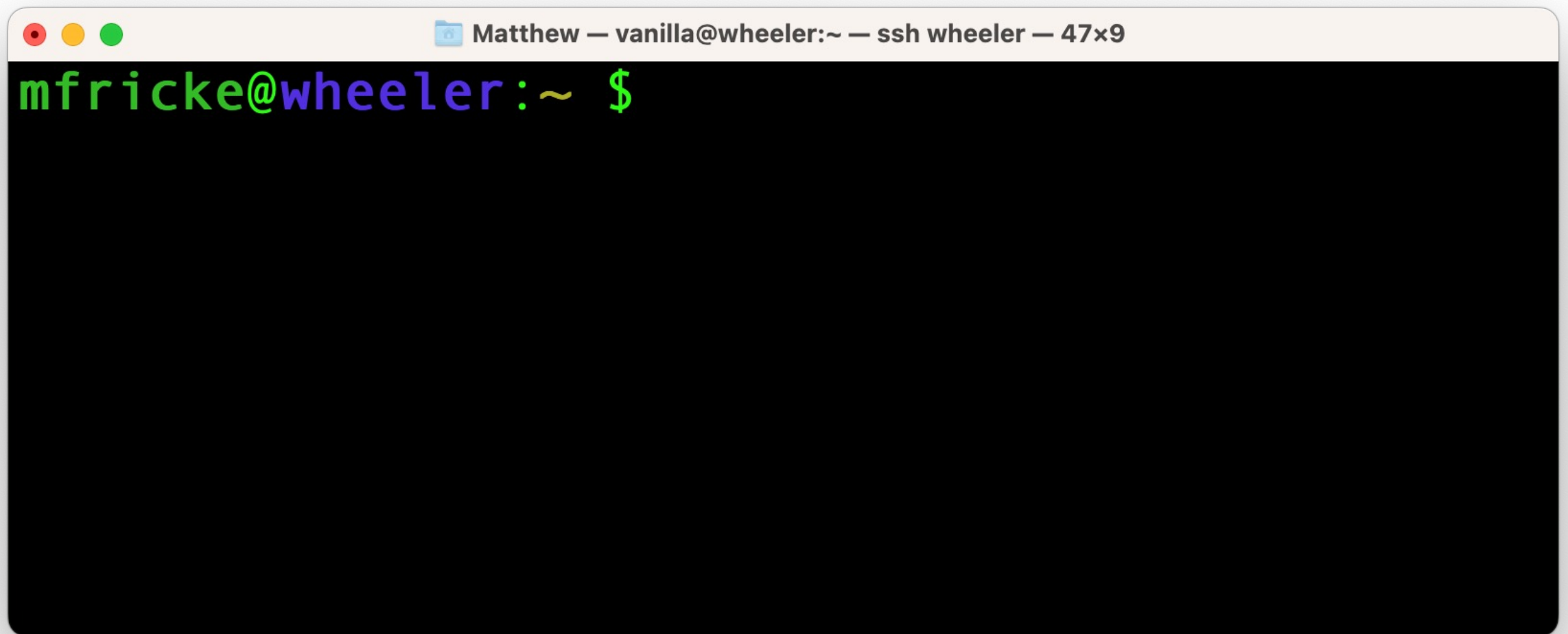
Linux and the BASH Shell

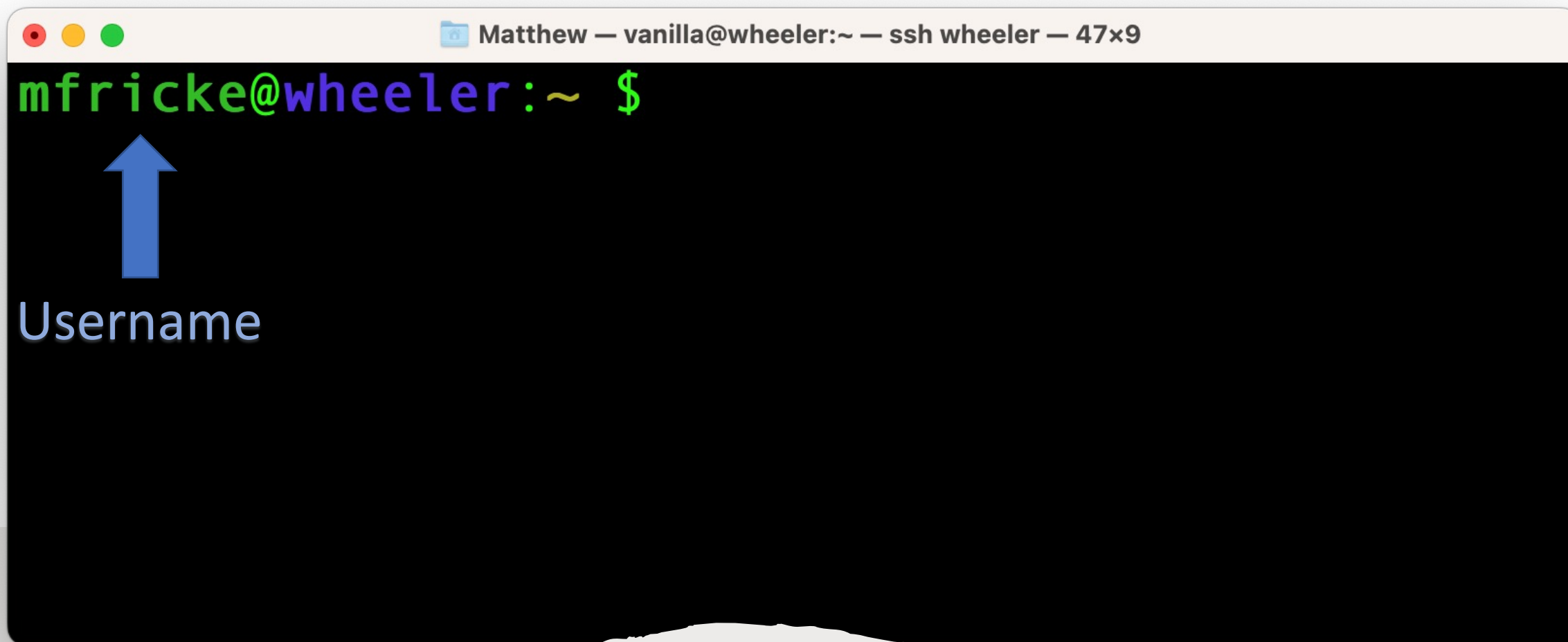
The word "LINUX" is rendered in a stylized, blocky font using only vertical and horizontal lines, giving it a digital or circuit-like appearance. Each letter is composed of multiple parallel lines, creating a sense of depth and texture. The 'L' is a simple vertical line with a horizontal base. The 'I' is a vertical line with a horizontal top bar. The 'N' is formed by a vertical line, a diagonal line, and a horizontal base. The 'U' is a vertical line with a horizontal base and a small vertical tick on the right side. The 'X' is formed by two intersecting diagonal lines with horizontal top and bottom bars.

WHERE THERE IS A SHELL, THERE IS A WAY!

The Borne-Again Shell (BASH)

Written in 1976 by Stephen Bourne for UNIX version 7.

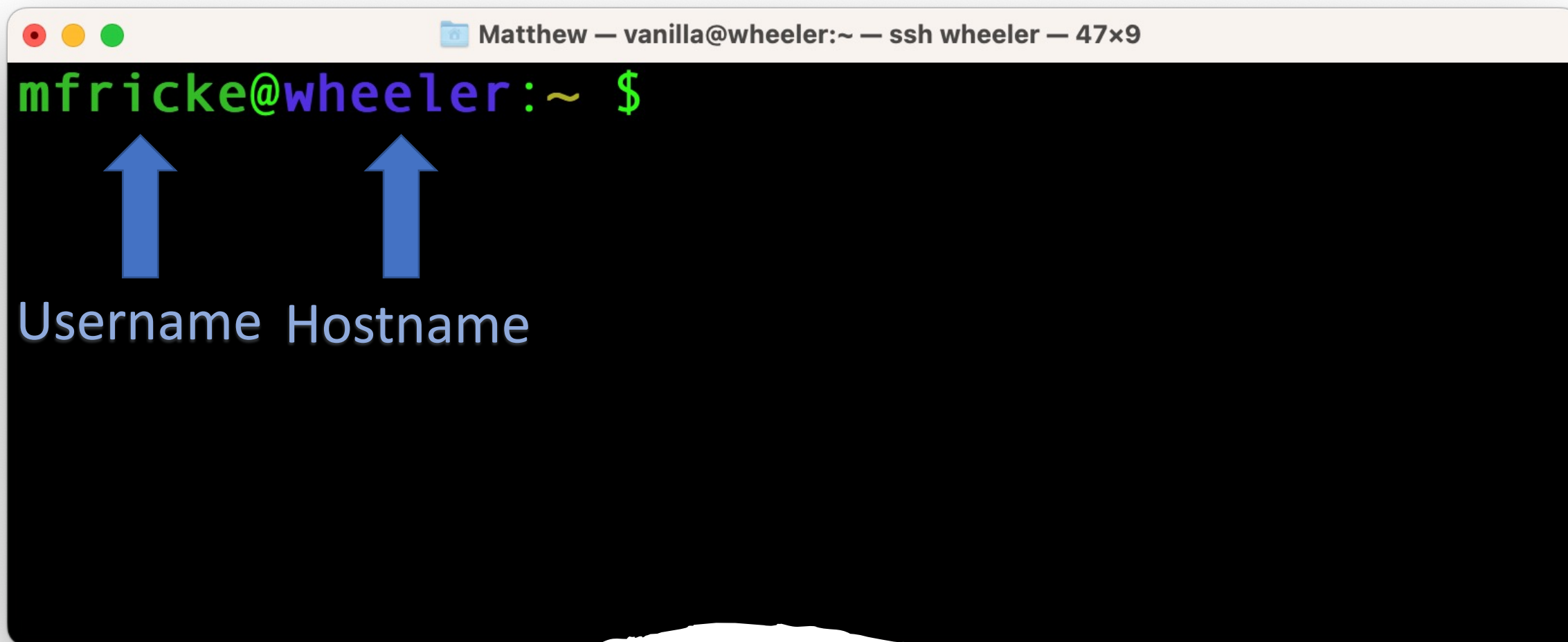




A terminal window with a title bar that reads "Matthew — vanilla@wheeler:~ — ssh wheeler — 47x9". The terminal content shows a BASH prompt "mfricke@wheeler:~ \$" where "mfricke" is green, "@wheeler" is blue, ":", "~" are yellow, and "\$" is green. A blue arrow points from the word "Username" below to the "mfricke" part of the prompt.

```
Matthew — vanilla@wheeler:~ — ssh wheeler — 47x9  
mfricke@wheeler:~ $  
↑  
Username
```

Understanding the BASH prompt...



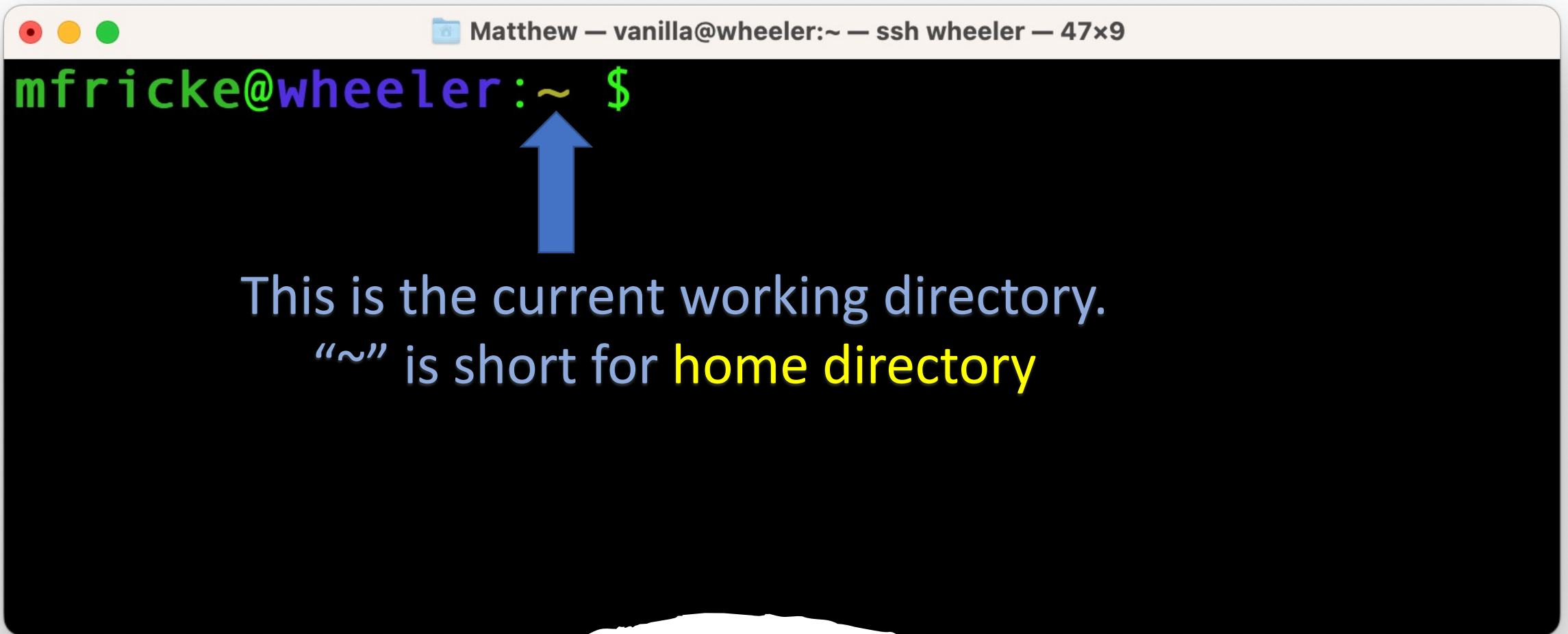
Matthew — vanilla@wheeler:~ — ssh wheeler — 47x9

```
mfricke@wheeler:~ $
```

↑ ↑

Username Hostname

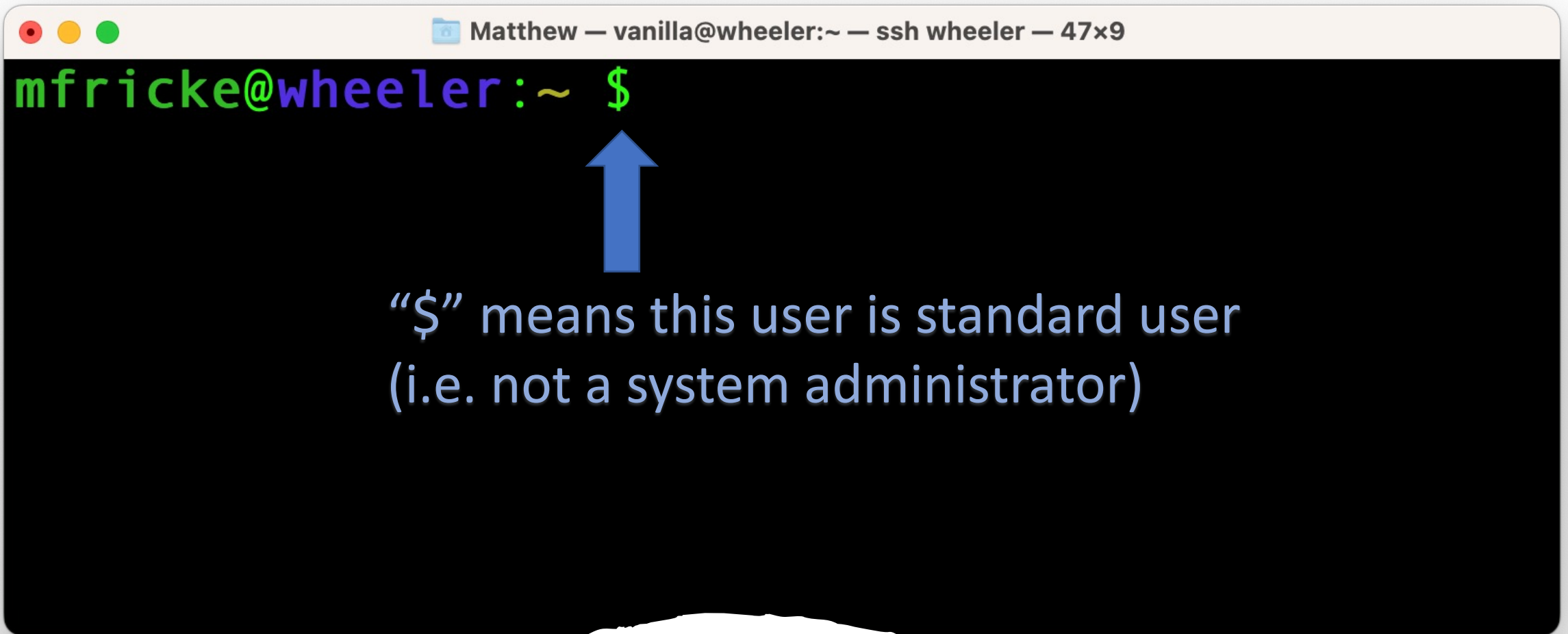
Understanding the BASH prompt...

A terminal window with a title bar that reads "Matthew — vanilla@wheeler:~ — ssh wheeler — 47x9". The terminal content shows the prompt "mfricke@wheeler:~ \$" in green and blue. A blue arrow points from the tilde character (~) to the text "This is the current working directory. '~' is short for home directory" which is displayed in light blue and yellow. The terminal background is black.

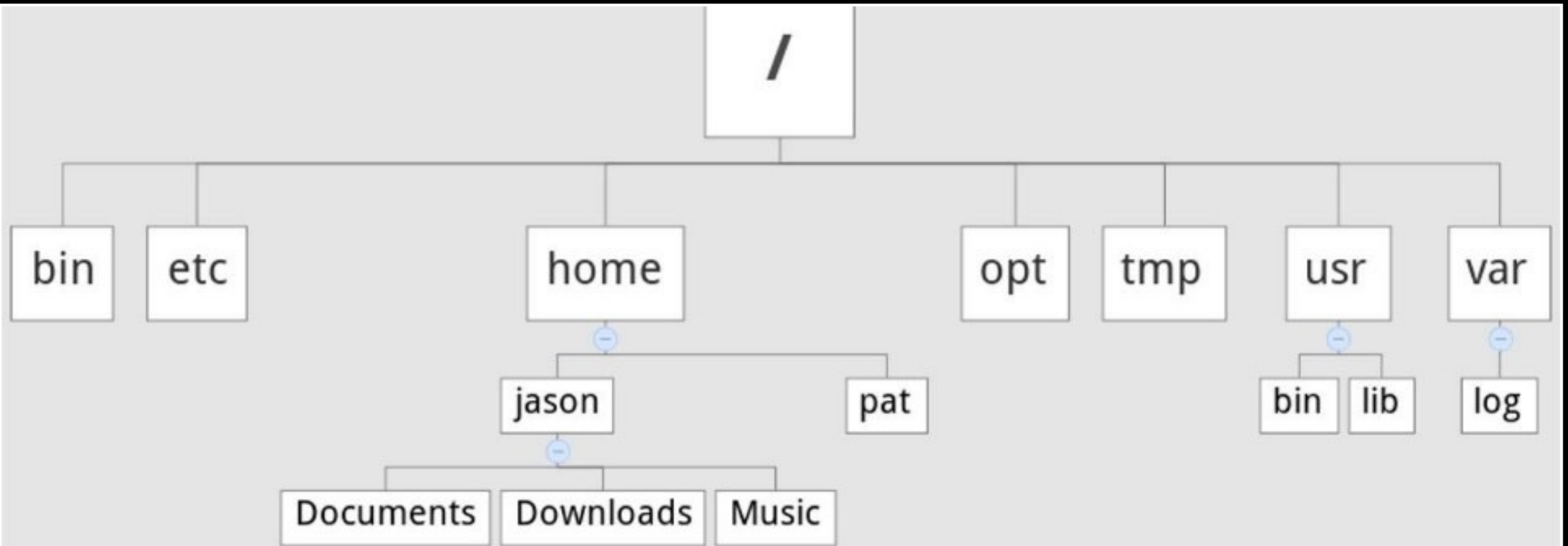
```
mfricke@wheeler:~ $
```

This is the current working directory.
“~” is short for **home directory**

Understanding the BASH prompt...



Understanding the BASH prompt...



Example Filesystem Tree


```
[vanilla@wheeler ~]$ pwd  
/users/vanilla  
[vanilla@wheeler ~]$
```

Figuring out where you are in the
filesystem...

```
[vanilla@wheeler ~]$ ls  
mystuff  wheeler-scratch  
[vanilla@wheeler ~]$
```

Figuring out where you are in the
filesystem...

```
[vanilla@wheeler ~]$ tree
```

```
.
├── mystuff
│   ├── myfile1
│   └── myfile2
└── wheeler-scratch -> /wheeler/scratch/vanilla
```

```
2 directories, 2 files
```

```
[vanilla@wheeler ~]$
```

Figuring out where you are in
the filesystem...

```
[vanilla@wheeler ~]$ tree
```



```
.  
├── mystuff  
│   ├── myfile1  
│   └── myfile2  
└── wheeler-scratch -> /wheeler/scratch/vanilla
```

This . means the current directory

```
2 directories, 2 files
```

```
[vanilla@wheeler ~]$
```

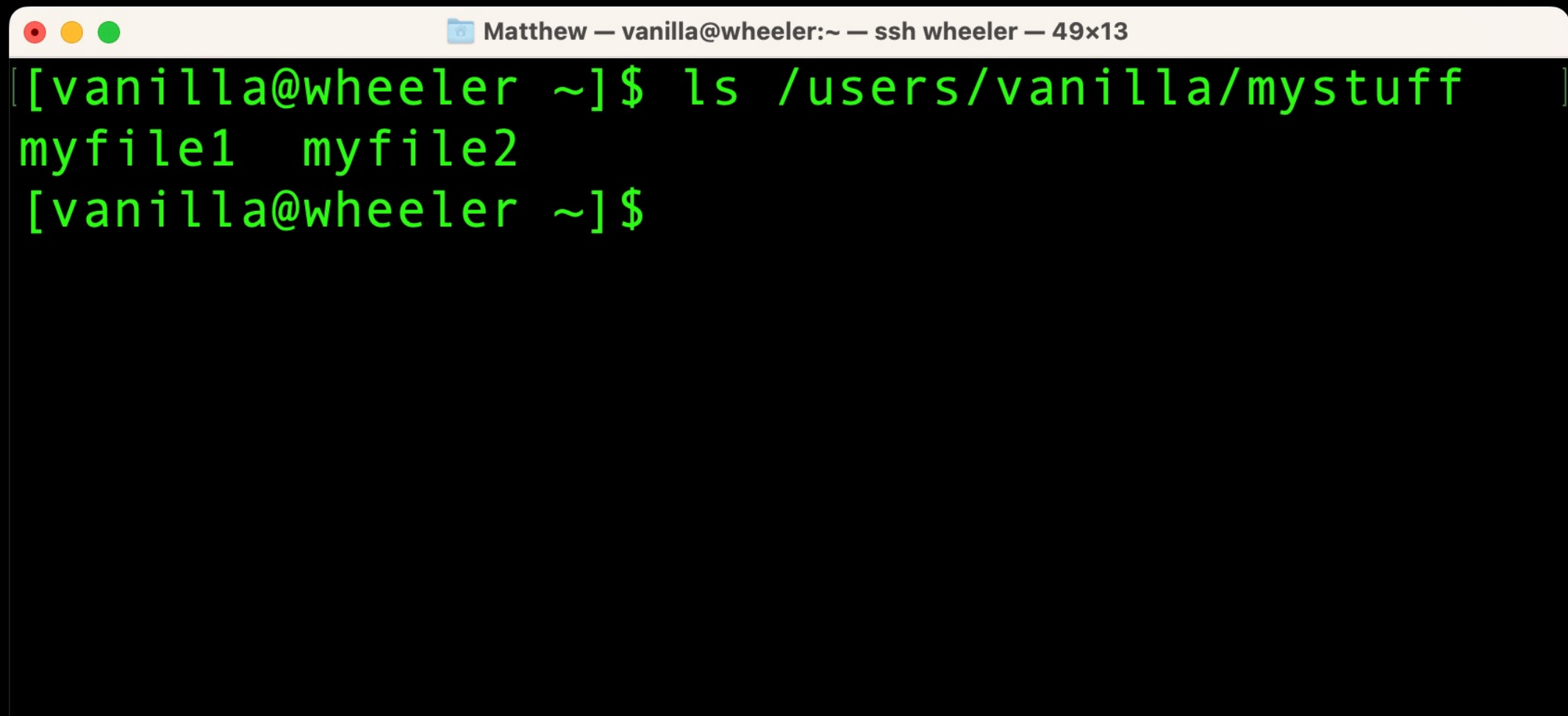
Figuring out where you are in
the filesystem...

“Absolute” paths vs “relative” paths

- A path is a list of directories and/or files. It is a path through the directory tree that tells one how to get somewhere in the filesystem.
- An absolute path tells one how to get to the destination from starting from the root of the filesystem. E.g “/users/vanilla/mystuff/”
- A relative path specifies how to get there *starting from the current working directory*. E.g vanilla/mystuff/

```
Matthew — vanilla@wheeler:~ — ssh wheeler — 49x13  
[vanilla@wheeler ~]$ ls mystuff/  
myfile1  myfile2  
[vanilla@wheeler ~]$
```

Figuring out where you going...

A terminal window with a title bar containing three colored circles (red, yellow, green) and the text "Matthew — vanilla@wheeler:~ — ssh wheeler — 49x13". The terminal content shows a command prompt "[vanilla@wheeler ~]\$", followed by the command "ls /users/vanilla/mystuff", and the output "myfile1 myfile2". The prompt then returns to "[vanilla@wheeler ~]\$".

```
Matthew — vanilla@wheeler:~ — ssh wheeler — 49x13  
[vanilla@wheeler ~]$ ls /users/vanilla/mystuff  
myfile1  myfile2  
[vanilla@wheeler ~]$
```

Figuring out where you going...

```
[vanilla@wheeler ~]$ ls ./mystuff/  
myfile1  myfile2  
[vanilla@wheeler ~]$ ls ~/mystuff/  
myfile1  myfile2  
[vanilla@wheeler ~]$
```

Figuring out where you going...


```
[vanilla@wheeler ~]$ ls -a
.          .modulesbeginenv
..         mystuff
.addressbook  .oracle_jre_usage
.addressbook.lu .pinerc
.bashrc      .pki
.cache       .rhosts
.comsol      .shosts
.config      .spack
.flexlmrc    .ssh
```

Figuring out where you going...

```
[vanilla@wheeler ~]$ ls -l
total 4
drwxr-xr-x 2 vanilla users 4096 Jun 14 22:05 mystuff
lrwxrwxrwx 1 vanilla users   24 Jun 14 21:20 wheeler-scratch ->
/wheeler/scratch/vanilla
[vanilla@wheeler ~]$
```

Figuring out where you going...

```
[vanilla@wheeler ~]$ ls -l mystuff/  
total 473704  
-rw-r--r-- 1 vanilla users 483165473 Jun 14 23:20 myfile1  
-rw-r--r-- 1 vanilla users          0 Jun 14 22:05 myfile2  
[vanilla@wheeler ~]$
```

Figuring out where you going...

```
[vanilla@wheeler ~]$ ls -lh mystuff/
total 463M
-rw-r--r-- 1 vanilla users 461M Jun 14 23:20 myfile1
-rw-r--r-- 1 vanilla users    0 Jun 14 22:05 myfile2
[vanilla@wheeler ~]$
```

Figuring out where you going...

```
[vanilla@wheeler ~]$ du -s  
499704 .  
[vanilla@wheeler ~]$ du -sh  
488M .  
[vanilla@wheeler ~]$
```

Figuring out where you going...

```
[vanilla@wheeler ~]$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	24G	0	24G	0%	/dev
tmpfs	24G	64K	24G	1%	/dev/shm
tmpfs	24G	968M	23G	5%	/run
tmpfs	24G	0	24G	0%	/sys/fs/cgroup
/dev/mapper/centos-root	930G	567G	363G	61%	/
/dev/sdc2	836G	72G	764G	9%	/tmp
/dev/md126p1	2.0G	333M	1.7G	17%	/boot
172.17.2.254:/mnt/wheeler-scratch	37T	28T	8.7T	77%	/wheeler/scratch
172.17.2.255:/mnt/wheeler-scratch2	37T	28T	9.0T	76%	/wheeler/scratch2
beegfs_nodev	110T	51T	60T	46%	/carc/scratch
chama:/home/homes	65T	36T	30T	55%	/users
chama:/home/carc_projects	65T	36T	30T	55%	/projects

Figuring out where you going...

```
[vanilla@wheeler ~]$ quota -s
Disk quotas for user vanilla (uid 659):
    Filesystem    space   quota   limit   grace   files   quota   limit   grace
chama:/home/homes
                488M    100G    200G           315   4295m   4295m
[vanilla@wheeler ~]$
```

Figuring out where you going...

```
[vanilla@wheeler ~]$ stat mystuff/myfile1
```

```
  File: 'mystuff/myfile1'  
  Size: 483165473  Blocks: 947408      IO Block: 65536   regular file  
Device: 28h/40d Inode: 9232782834205560540  Links: 1  
Access: (0644/-rw-r--r--)  Uid: ( 659/  vanilla)   Gid: ( 100/   users)  
Access: 2022-06-14 22:05:27.503289000 -0600  
Modify: 2022-06-14 23:20:26.945918000 -0600  
Change: 2022-06-14 23:20:48.754917000 -0600  
  Birth: -
```

Figuring out what you've got...


```
[vanilla@wheeler ~]$ find -name myfile2  
./mystuff/myfile2
```

```
[vanilla@wheeler ~]$ find -name "myfile*"  
./mystuff/myfile1  
./mystuff/myfile2  
./mystuff/myfile3  
./mystuff/myfile0
```



Wildcard

Figuring out what you've got...

```
[vanilla@wheeler ~]$ cd mystuff/  
[vanilla@wheeler ~/mystuff]$
```

Use the tab key to autocomplete

Going somewhere new...

Now it is your turn...



- For this path:
`/projects/shared/workshops/beginner/vecadd`

- What are the names of the files in that directory?
- When were they last modified?
- How large are the files?

You can find this information with the `ls` command.

Now it is your turn...



- For this path:
`/projects/shared/workshops/beginner/vecadd`

Now “cd” into that directory using <tab> autocomplete.

- Now you know how to find your way around filesystems using bash
- Let's see how to modify the filesystem.
- In bash to move a file we use the `mv` command.
- To copy a file it is `cp`.
- To copy files from CARC to a personal computer use `scp` or `rsync`.

```
[vanilla@wheeler beginner]$ pwd
/projects/shared/workshops/beginner
[vanilla@wheeler beginner]$ cd ~
[vanilla@wheeler ~]$ pwd
/users/vanilla
[vanilla@wheeler ~]$
```

First return to your home
directory...

```
[vanilla@wheeler ~]$ cd mystuff
[vanilla@wheeler ~/mystuff]$ mv myfile1 myfile0
[vanilla@wheeler ~/mystuff]$ ls
myfile0  myfile2  myfile3
[vanilla@wheeler ~/mystuff]
```

Modifying the filesystem...
moving a file.

```
[vanilla@wheeler ~/mystuff]$ cp myfile0 myfile1  
[vanilla@wheeler ~/mystuff]$
```



Source



Destination

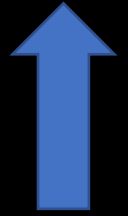
```
[vanilla@wheeler ~/mystuff]$ ls  
myfile0  myfile1  myfile2  myfile3  
[vanilla@wheeler ~/mystuff]$
```

Modifying the filesystem...
copying a file.


```
[vanilla@wheeler ~]$ cp -r mystuff mystuff2  
[vanilla@wheeler ~]$
```



Source



Destination

```
[vanilla@wheeler ~]$ ls  
mystuff    mystuff2    wheeler-scratch
```

Copying a whole directory tree...

```
Lycaon:~ matthew$ scp vanilla@wheeler.alliance.unm.edu:~/mystuff/myfile3 Desktop/
```



Source



Destination

```
(vanilla@wheeler.alliance.unm.edu) Password:  
myfile3          100%  40      2.0KB/s   00:00
```

Copying data to a personal
computer from CARC...

```
Lycaon:~ matthew$ scp -r vanilla@wheeler.alliance.unm.edu:~/mystuff Desktop/
```



Source



Destination

```
(vanilla@wheeler.alliance.unm.edu) Password:
```

myfile1	100%	1024KB	6.5MB/s	00:00
myfile2	100%	2048KB	382.5KB/s	00:05
myfile3	100%	40	3.2KB/s	00:00
myfile0	100%	1024KB	8.8MB/s	00:00

Copying data to a personal
computer from CARC...

```
Lycaon:~ matthew$ scp -r Desktop/mystuff vanilla@wheeler.alliance.unm.edu:~/
```



Source



Destination

```
(vanilla@wheeler.alliance.unm.edu) Password:
```

myfile1	100%	1024KB	591.5KB/s	00:01
myfile0	100%	1024KB	2.0MB/s	00:00
myfile2	100%	2048KB	2.1MB/s	00:00
myfile3	100%	40	2.1KB/s	00:00

To copy from a personal
computer to CARC...

```
ssh vanilla@wheeler.alliance.unm.edu
```

Log back into wheeler...

```
[vanilla@wheeler ~]$ file mystuff/myfile0  
mystuff/myfile0: data
```

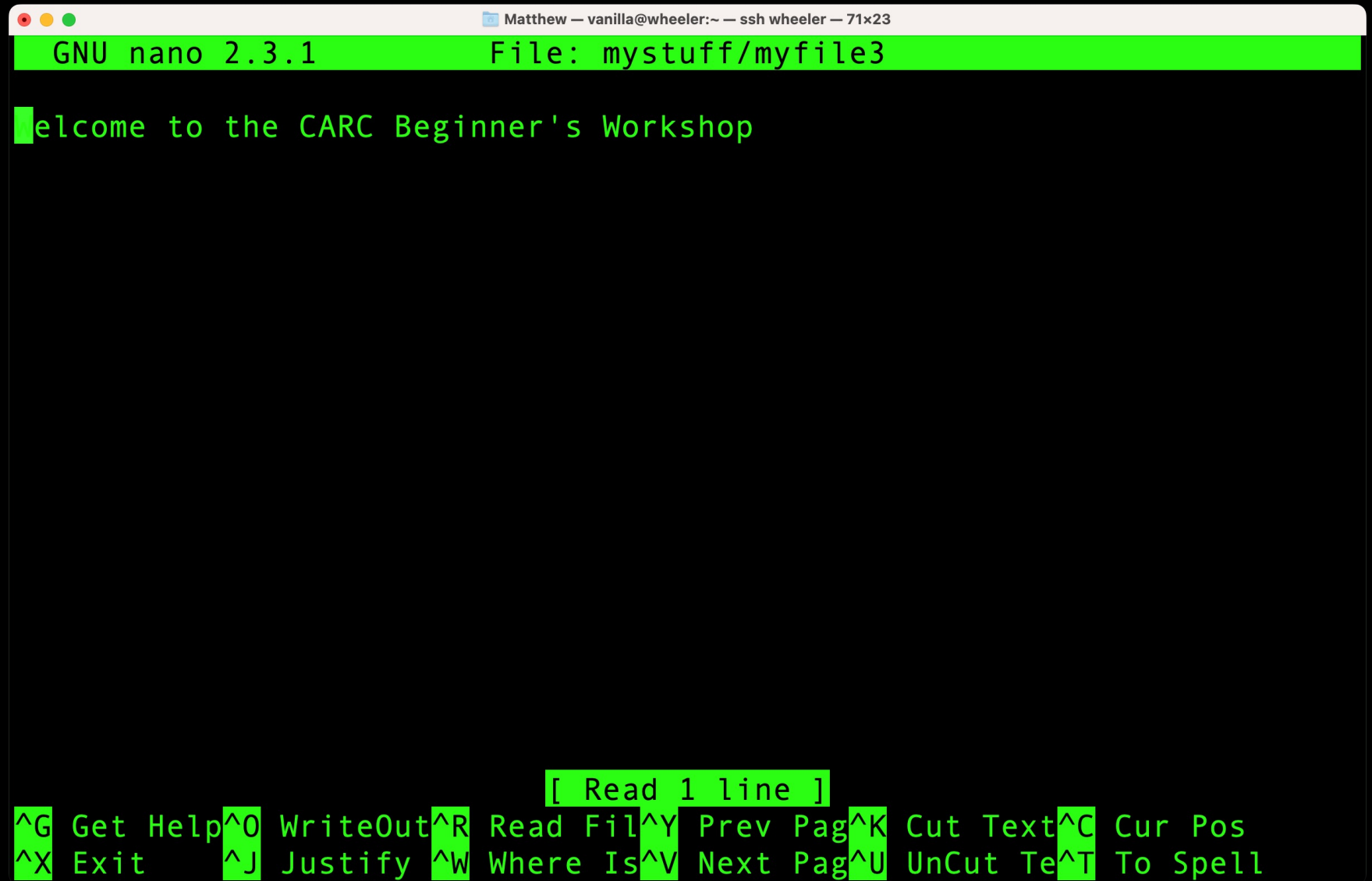
```
[vanilla@wheeler ~]$ file mystuff/myfile3  
mystuff/myfile3: ASCII text
```

Figuring out file types ...

```
[vanilla@wheeler ~]$ cat mystuff/myfile3  
Welcome to the CARC Beginner's Workshop
```

Text files ...

```
[vanilla@wheeler ~]$ nano mystuff/myfile3
```

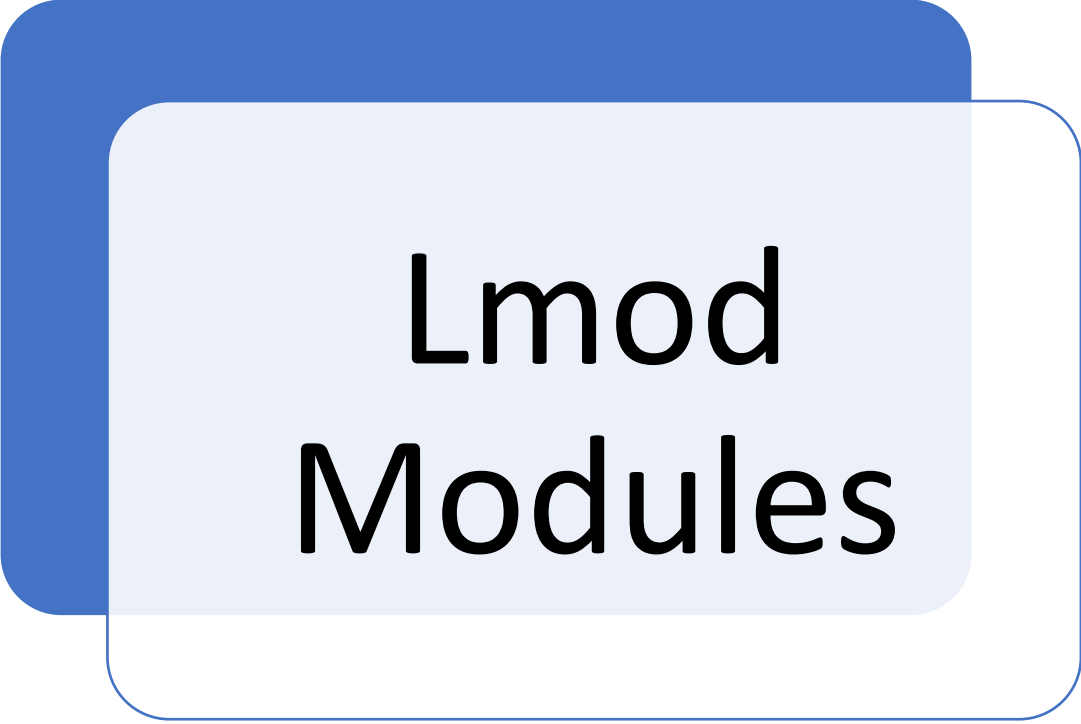


```
Matthew — vanilla@wheeler:~ — ssh wheeler — 71x23
GNU nano 2.3.1 File: mystuff/myfile3

Welcome to the CARC Beginner's Workshop

[ Read 1 line ]
^G Get Help ^O WriteOut ^R Read Fil ^Y Prev Pag ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Pag ^U UnCut Te ^T To Spell
```


Software Access



Lmod
Modules

The diagram consists of a dark blue rounded rectangle with a light blue rounded rectangle nested inside it. The text "Lmod Modules" is centered within the light blue rectangle.



Conda

The diagram consists of a dark blue rounded rectangle with a light blue rounded rectangle nested inside it. The text "Conda" is centered within the light blue rectangle.

```
[vanilla@wheeler ~]$ module spider matlab
```

```
-----
```

```
-
```

```
  matlab:
```

```
-----
```

```
-
```

```
  Versions:
```

```
    matlab/R2017a
```

```
    matlab/R2018b
```

```
    matlab/R2019a
```

```
    matlab/R2020a
```

```
    matlab/R2021a
```

Getting access to software...

```
[vanilla@wheeler ~]$ module load matlab/R2021a
Lmod has detected the following error: Matlab may only be run on compute
nodes. wheeler is not a compute node. Exiting...
While processing the following module(s):
  Module fullname  Module Filename
  -----
matlab/R2021a      /opt/local/modules/matlab/R2021a.lua
```

Getting access to software...

```
[vanilla@wheeler ~]$ module load matlab/R2021a
Lmod has detected the following error: Matlab may only be run on compute
nodes. wheeler is not a compute node. Exiting...
While processing the following module(s):
  Module fullname  Module Filename
  -----
matlab/R2021a      /opt/local/modules/matlab/R2021a.lua
```

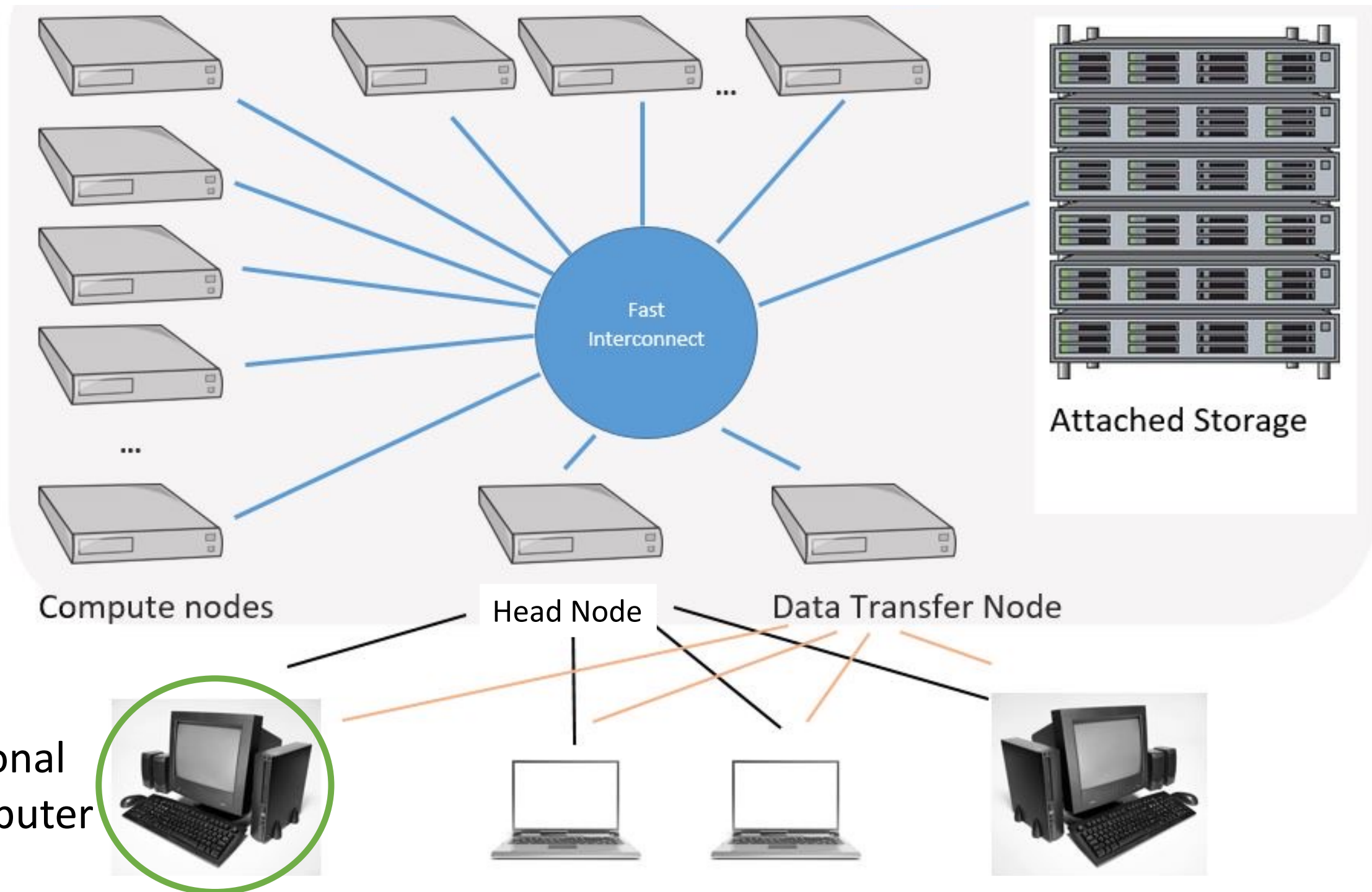
What is a compute node?

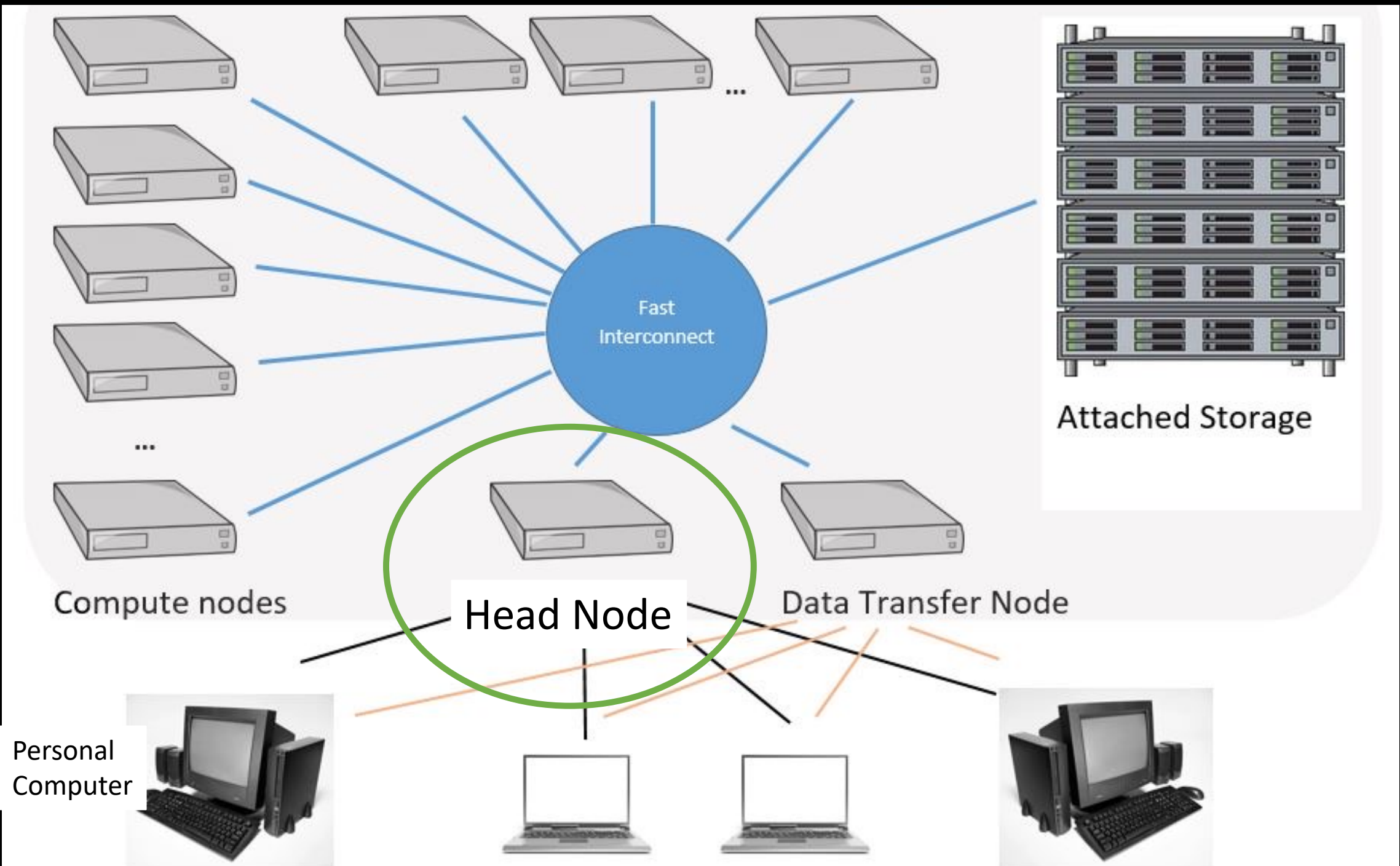
Getting access to software...

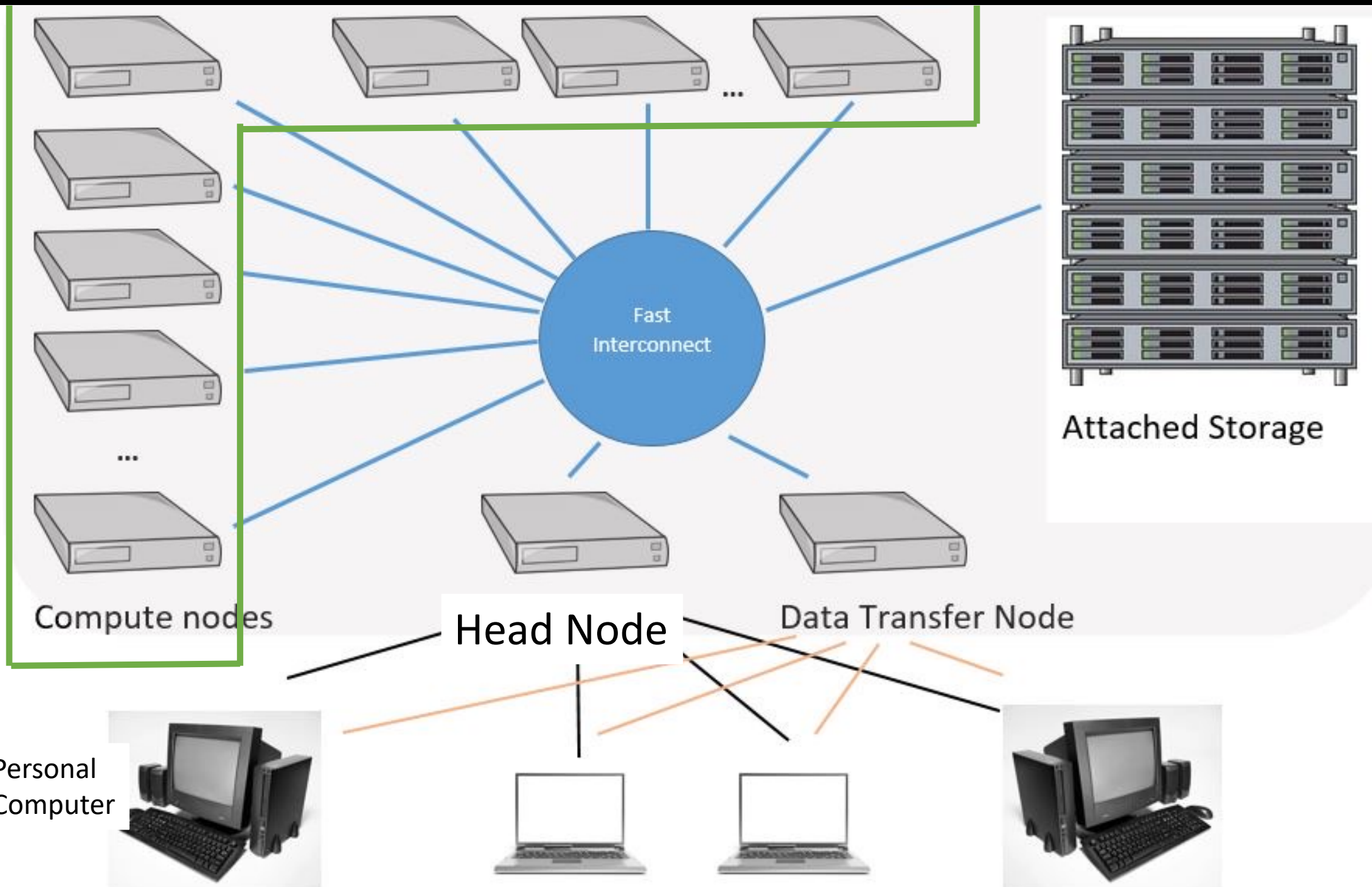
HPC Cluster



Personal
Computer









Never run computations on the head node

Always use compute nodes

```
[vanilla@wheeler ~]$ qgrok
```

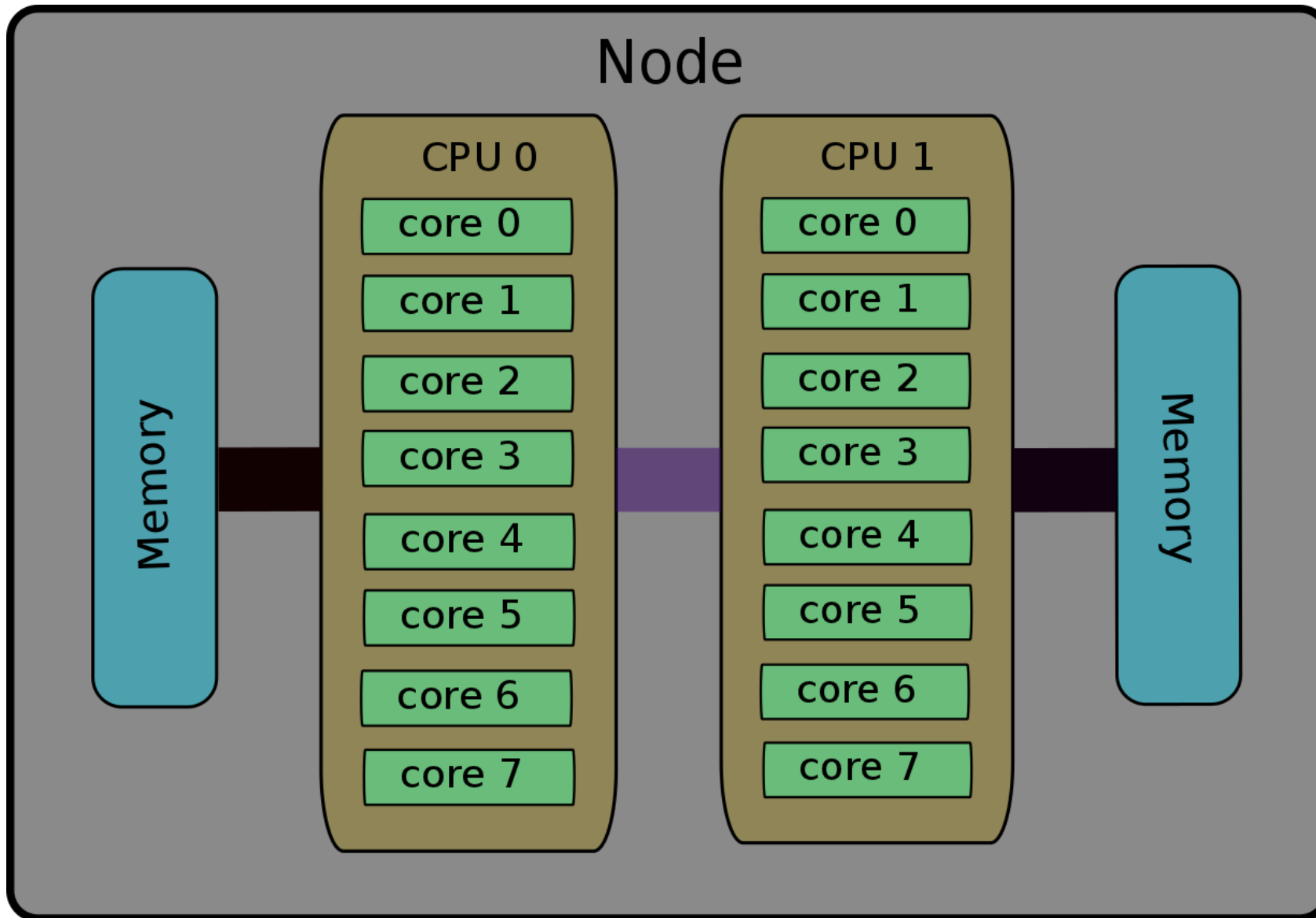
queues	free	busy	offline	jobs	nodes	CPUs
-----	-----	-----	-----	-----	-----	-----
normal	0	299	1	97	300	2400
debug	4	0	0	0	4	32
totals:	4	299	1	97	304	2432

Compute nodes and partitions...

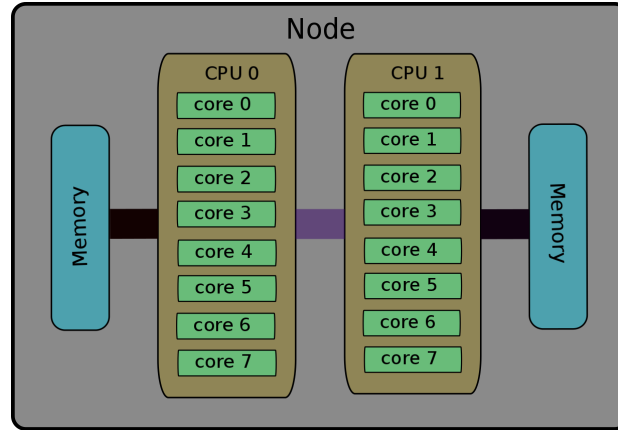
```
qmfricke@hopper:~ $ qgrok
```

queues	free	busy	offline	jobs	nodes	CPUs
-----	-----	-----	-----	-----	-----	-----
general	12	0	0	0	12	384
condo	29	1	7	0	37	1184
bugs	2	0	0	0	2	64
pcnc	2	0	0	0	2	64
pathogen	0	1	0	1	1	32
tc	7	0	3	0	10	320
gold	2	0	0	0	2	64
fishgen	1	0	0	0	1	32
neuro-hsc	13	0	1	0	14	448
cup-ecs	2	0	0	0	2	64
tid	0	0	1	0	1	32
biocomp	0	0	1	0	1	32
chakra	0	0	1	0	1	32
totals:	70	2	14	1	86	2752

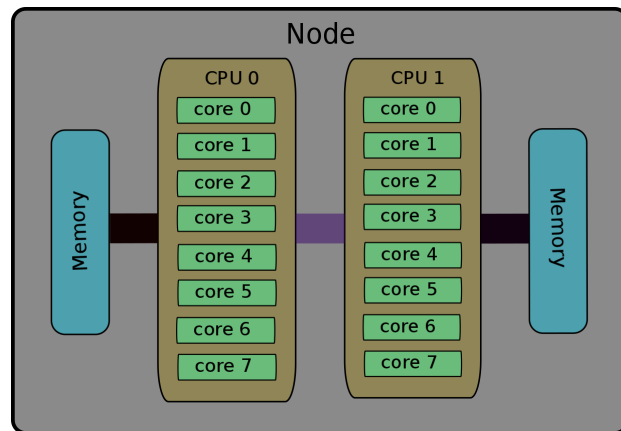
Compute nodes and partitions...



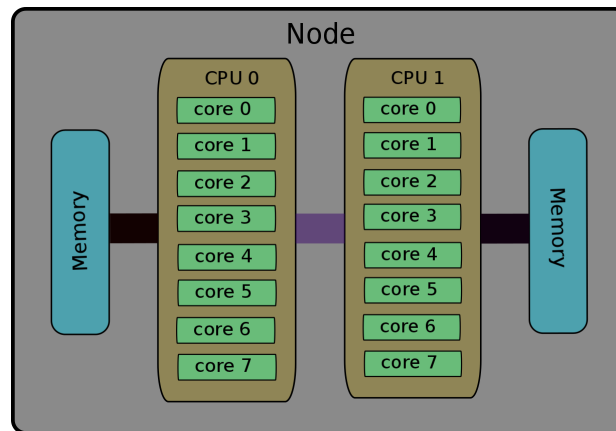
Head Node (wheeler)



Compute Nodes

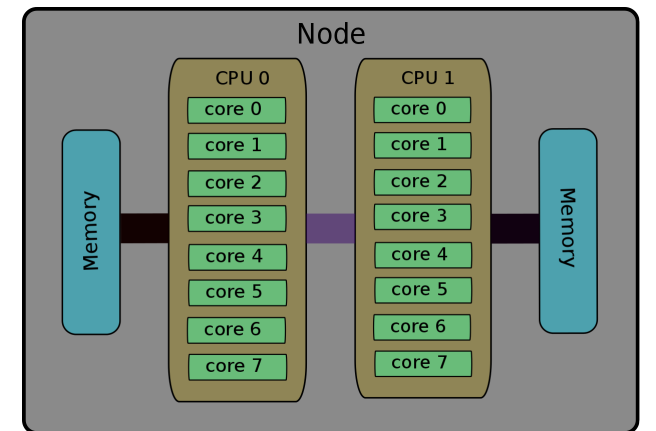


wheeler001



wheeler002

...



wheeler304

```
[vanilla@wheeler ~]$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Byte Order:            Little Endian
CPU(s):                8
On-line CPU(s) list:   0-7
Thread(s) per core:    1
Core(s) per socket:    4
Socket(s):             2
NUMA node(s):          2
Vendor ID:             GenuineIntel
CPU family:            6
Model:                26
Model name:            Intel(R) Xeon(R) CPU X5550  @ 2.67GHz
```

Wheeler has 8 cores per node....

```
mfricke@hopper:~ $ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Byte Order:            Little Endian
CPU(s):                64
On-line CPU(s) list:   0-63
Thread(s) per core:    2
Core(s) per socket:    16
Socket(s):             2
NUMA node(s):          2
Vendor ID:             GenuineIntel
CPU family:            6
Model:                 85
Model name:            Intel(R) Xeon(R) Gold 6226R CPU @ 2.90GHz
```

Hopper has 32 cpus per node....

Architecture

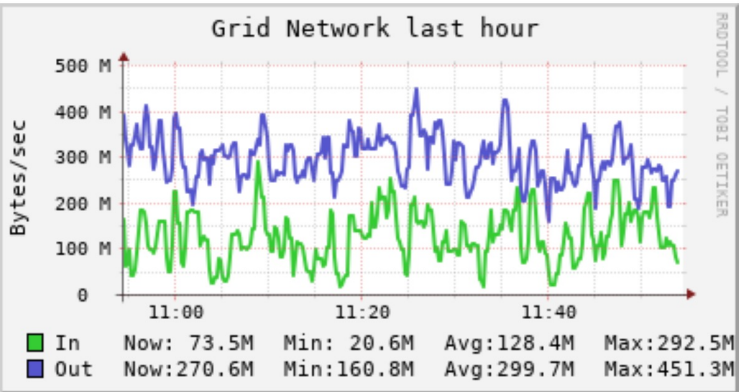
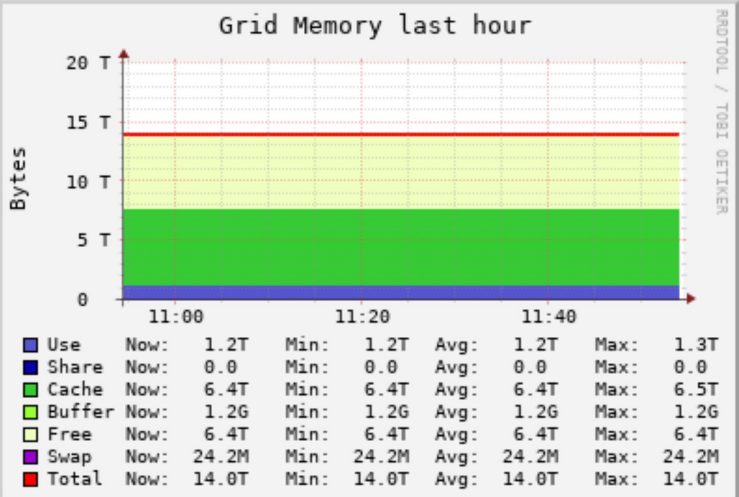
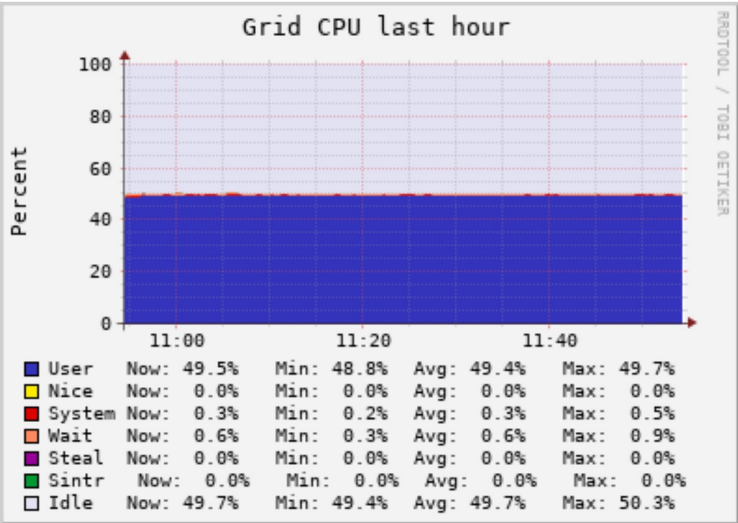
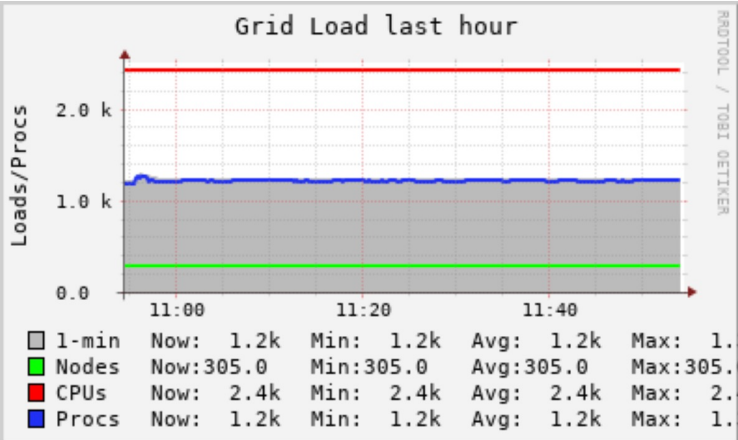
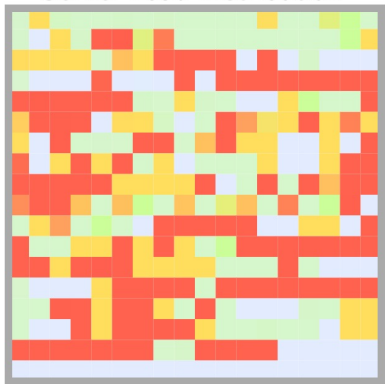
Wheeler Cluster Grid > Wheeler Cluster > --Choose a Node

Overview of @ 2022-03-03 18:54

CPU's Total: 2440
Hosts up: 305
Hosts down: 0

Current Load Avg (15, 5, 1m):
51%, 51%, 51%
Avg Utilization (last hour):
51%

Server Load Distribution





Technology, IT etc.

SLURM

means

Simple Linux Utility for Resource
Management

by acronymsandslang.com



VOTED **#1** SOFT DRINK OF THE 31ST CENTURY!



```
[vanilla@wheeler ~]$ squeue
```

	JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
	159914	normal	co-mcpdf	nsharma2	CG	2-00:00:26	1	wheeler257
	159915	normal	co-mcpdf	nsharma2	CG	2-00:00:26	1	wheeler257
	159916	normal	co-mcpdf	nsharma2	CG	2-00:00:26	1	wheeler257
	159917	normal	co-mcpdf	nsharma2	CG	2-00:00:26	1	wheeler257
	159918	normal	co-mcpdf	nsharma2	CG	2-00:00:26	1	wheeler257
	159919	normal	co-mcpdf	nsharma2	CG	2-00:00:26	1	wheeler257
	159912	normal	co-mcpdf	nsharma2	CG	2-00:00:28	1	wheeler257
	159913	normal	co-mcpdf	nsharma2	CG	2-00:00:28	1	wheeler257
166800_[21-100%10]		normal	Jannat	jannat	PD	0:00	1	(JobArrayTaskLimit)
	167067	normal	WINDENER	rubeldas	PD	0:00	36	
(QOSMaxCpuPerUserLimit)								
	167068	normal	WINDENER	rubeldas	PD	0:00	24	

Slurm....

```
[vanilla@wheeler ~]$ srun --partition debug --nodes 2 hostname  
srun: Account not specified in script or  
~/.default_slurm_account, using latest project  
wheeler302.alliance.unm.edu  
You have not been allocated GPUs. To request GPUs, use the -G  
option in your submission script.  
wheeler301.alliance.unm.edu  
[vanilla@wheeler ~]$ srun --partition debug --nodes 2 hostname
```

The srun command...

```
[vanilla@wheeler ~]$ srun --partition debug --ntasks 8 hostname
srun: Account not specified in script or ~/.default_slurm_account, using
latest project
wheeler302.alliance.unm.edu
wheeler302.alliance.unm.edu
wheeler302.alliance.unm.edu
wheeler302.alliance.unm.edu
wheeler302.alliance.unm.edu
wheeler302.alliance.unm.edu
wheeler302.alliance.unm.edu
You have not been allocated GPUs. To request GPUs, use the -G option in your
submission script.
wheeler302.alliance.unm.edu
wheeler302.alliance.unm.edu
```

The srun command...

JupyterHub



python™

julia



Welcome to the

This Notebook Server was

WARNING

Don't rely on this serv

Your server is hosted thar

Run some Python

To run the code below:

1. Click on the cell to se
2. Press SHIFT+ENTER

A full tutorial for using the

In []: `matplotlib inline`

```
import pandas as pd
import numpy as np
import matplotlib
```

Exploring the Lorenz System

In this Notebook we explore the [Lorenz system](#) of differential equations:

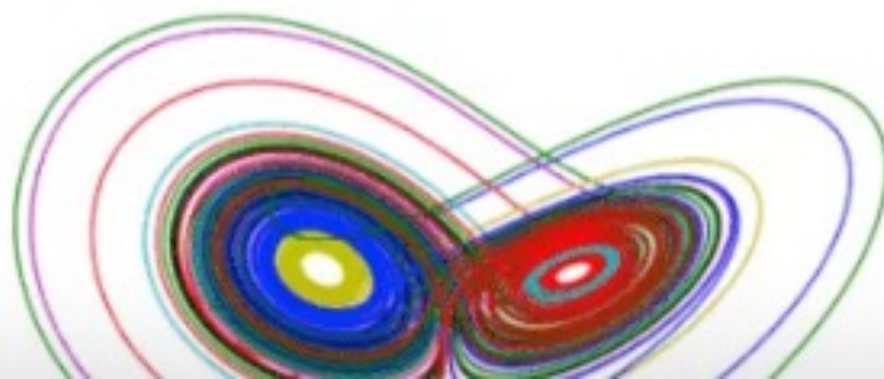
$$\dot{x} = \sigma(y - x)$$

$$\dot{y} = \rho x - y - xz$$

$$\dot{z} = -\beta z + xy$$

This is one of the classic systems in non-linear differential equations. It exhibits a range of complex behaviors as the parameters (σ, β, ρ) are varied, including what are known as chaotic solutions. The system was originally developed as a simplified mathematical model for atmospheric convection in 1963.

```
In [7]: interact(Lorenz, N=fixed(10), angle=(0.,360.),
                 sigma=(0.0,50.0), beta=(0.,5), rho=(0.0,50.0))
```



The Hopper Cluster

- Login to hopper.
- If you are already logged into wheeler this is just:

```
ssh hopper
```

If you are not then

```
ssh username@hopper.alliance.unm.edu
```


https://github.com/unm-escape/okada_jupyter

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Notifications

<>Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

main1 branch0 tags

Go to fileCode

About

ericlindsey Update README.md

AlaskaFaultsShapefiles	Initial commit of files	
Surface_Rupture_Ridgecrest_Prov...	Initial commit of files	
GPS_Alaska2021.csv	Initial commit of files	
GPS_Ridgecrest.csv	Initial commit of files	
README.md	Update README.md	
fault_model.py	Initial commit of files	
fault_plots.py	Initial commit of files	4 days ago
fit_fault_model.ipynb	Initial commit of files	4 days ago
geod_transform.py	Initial commit of files	4 days ago
moment_tensor.py	Initial commit of files	4 days ago
okada85.py	Initial commit of files	4 days ago
okada_greens.py	Initial commit of files	4 days ago

Clone

HTTPSGitHub CLIGo to fileCode

https://github.com/unm-escape/okada_ju

Use Git or checkout with SVN using the web URL.

Open with GitHub Desktop

Download ZIP

README.md

Jupyterlab example for Okada fault modeling

About

Okada fault modeling to fit GPS data in Jupyter

Readme

1 star

1 watching

1 fork

Releases

No releases published

Packages

No packages published

Languages

Python 81.5%

Jupyter Notebook 18.5%

main1 branch0 tags

Go to fileCode

	ericlindsey Update README.md	
	AlaskaFaultsShapefiles	Initial commit of files
	Surface_Rupture_Ridgecrest_Prov...	Initial commit of files
	GPS_Alaska2021.csv	Initial commit of files
	GPS_Ridgecrest.csv	Initial commit of files
	README.md	Update README.md
	fault_model.py	Initial commit of files
	fault_plots.py	Initial commit of files4 days ago
	fit_fault_model.ipynb	Initial commit of files4 days ago
	geod_transform.py	Initial commit of files4 days ago
	moment_tensor.py	Initial commit of files4 days ago
	okada85.py	Initial commit of files4 days ago
	okada_greens.py	Initial commit of files4 days ago

Clone

HTTPSGitHub CLIGitHub CLIhttps://github.com/unm-escape/okada_juhttps://github.com/unm-escape/okada_juUse Git or checkout with SVN using the web URL.

Open with GitHub Desktop

Download ZIP

About

Okada fault modeling to fit GPS data in Jupyter

Readme

1 star

1 watching

1 fork

Releases

No releases published

Packages

No packages published

Languages

Python81.5%

Jupyter Notebook18.5%

README.md

Jupyterlab example for Okada fault modeling

```
[vanilla@hopper ~]$ git clone https://lobogit.unm.edu/CARC/workshops.git
Cloning into 'workshops'...
remote: Enumerating objects: 132, done.
remote: Counting objects: 100% (75/75), done.
remote: Compressing objects: 100% (43/43), done.
remote: Total 132 (delta 33), reused 74 (delta 32), pack-reused 57
Receiving objects: 100% (132/132), 57.58 KiB | 3.60 MiB/s, done.
Resolving deltas: 100% (51/51), done.
```

**We will use a popular tool called
“git” to download examples...**

Dependencies

- The example code depends on these libraries:
 - numpy
 - matplotlib
 - H5py
 - Scipy
 - simplekml
-
- Using it with jupyterhub requires [ipykernel](#) and [jupyter_client](#)

```
vanilla@hopper:~ $ module load miniconda3  
vanilla@hopper:~ $ conda create --name escape numpy  
scipy jupyter_client ipykernel h5py matplotlib  
simplekml --channel conda-forge
```

```
Collecting package metadata (current_repodata.json): done
```

```
Solving environment: done
```

```
## Package Plan ##
```

```
environment location: /users/mfricke/.conda/envs/fit_fault_model
```

```
Proceed ([y]/n)?
```

Downloading and Extracting Packages

```
entrypoints-0.4          | 16 KB          |  
##### | 100%
```

```
bottleneck-1.3.5        | 274 KB         |  
##### | 100%
```

[SNIP]

```
broccoli-1.0.9           | 19 KB          |  
##### | 100%
```

Preparing transaction: done

Verifying transaction: done

Executing transaction: done

#

To activate this environment, use

#

\$ conda activate escape

#

To deactivate an active environment, use

#

\$ conda deactivate

vanilla@hopper:~ \$

Welcome to the Center for Advanced Research Computing

The UNM Center for Advanced Research Computing is the hub of computational research at UNM and one of the largest computing centers in the State of New Mexico. It is an interdisciplinary community that uses computational resources to create new research insights. The goal is to lead and grow the computational research community at UNM.

CARC provides not just the computing resources but also the expertise and support to help the university's researchers. This service is available to faculty, staff, and student researchers free of charge through support from the UNM Office of the Vice President for Research.

Get sta

Create a help ticket

Downtime notices

CARC systems information

CARC infrastructure

Resource limits

Storage and backup

Export control

JupyterHub cluster links

System status and downtime

System Usage by Principal Investigator

 Hopper

Gibbs

Taos

Wheeler

Xena

JupyterHub cluster links

Students go on-site to study greenhouse gas emissions

UNM Earth Science Professor to study the health of groundwater aquifers across the state

Explore or exploit: How our brains make choices



Sign in

Username:

Password:

Sign in

Server Options

Select a job profile:

- ✓ Debug Queue, 1 hours, 1 core, 4GB RAM
- Debug Queue, 4 hours, 8 cores, 24GB RAM
- General Access Queue, 48 hours, 32 cores, 90GB RAM
- Condo Queue (Preemptable), 48 hours, 32 cores, 90GB RAM
- Condo Queue (Preemptable), 48 hours, 32 cores, 90GB RAM, 1 GPU
- Chakra (restricted), 48 hours, 32 cores, 180GB RAM, 1 GPU
- Geodef (restricted), 1 hours, 1 core, 14GB RAM
- Geodef (restricted), 48 hours, 32 cores, 500GB RAM



You will be redirected automatically when it's ready for you.

Event log

Files

Running

Clusters

Select items to perform actions on them.

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☐	0 ▾	 /		
☐		carc-scratch		
☐		mystuff		
☐		simcov		
☐		vecadd		
☐		wheeler-scratch	a year ago	
☐		workshops	11 minutes ago	
☐		hopper-scratch	a year ago	23 B
☐		jupyterhub_wheeler_batchspawner_552957.log	7 minutes ago	85 B
☐		jupyterhub_wheeler_batchspawner_552957.log	2 minutes ago	85 B

Notebook:

Python 3

Python [conda env:.conda-escape]

Other:

Text File

Folder

Terminal

[Logout](#)[Control Panel](#)[File](#) [Edit](#) [View](#) [Insert](#) [Cell](#) [Kernel](#) [Widgets](#) [Help](#)[Trusted](#)[Python 3](#) 

Run



Code



```
In [1]: 1+1
```

```
Out[1]: 2
```

```
In [2]: print("Hello World!")
```

```
Hello World!
```

```
In [ ]: |
```

Press shift-enter together to execute the current cell



Logout

Control Panel

File Edit View Insert Cell Kernel Widgets Help

Trusted



Python 3 



Run



Code



In [1]: 1+1

Out[1]: 2

In [2]: print("Hello World!")

Hello World!

In []: |

Click here to go back to the default screen

Select items to perform actions on them.

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[New ▼](#)


0



/

Name ▼

Last Modified

File size



carc-scratch

a year ago



mystuff

a year ago



simcov

7 months ago



vecadd

a year ago



wheeler-scratch

a year ago



workshops

seconds ago



hopper-scratch

a year ago

23 B



jupyterhub_wheeler_batchspawner_552957.log

22 minutes ago

0 B



jupyterhub_wheeler_slurm_cmd.err

in a few seconds

5.57 kB



jupyterhub_wheeler_slurm_cmd.out

22 minutes ago

0 B

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[Clusters](#)

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☐ 0 ▾	 / workshops	Name ▾	Last Modified	File size
☐  ..			seconds ago	
☐  escape			a minute ago	
☐  intro_workshop			a minute ago	
☐  namd			a minute ago	
☐  radio_astronomy			a minute ago	
☐  README.md			a minute ago	47 B

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[Clusters](#)

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0


📁 / workshops / escape

Name ▼

Last Modified

File size



..

seconds ago



fit_insar_data.ipynb



2 minutes ago

14 MB



kml_functions.py

2 minutes ago

4.41 kB



Kernel starting, please wait...

Not Trusted

Python [conda env:.conda-escape] ●

File Edit View Insert Cell Kernel Widgets Help



Run



Markdown



Viewing and interacting with a large InSAR dataset on CARC

The InSAR dataset "geo_timeseries_ERA5_ramp_demErr_msk.h5" contains the complete time-series of deformation between 2014 and 2023 over the city of Jakarta, Indonesia. The file "geo_velocity_msk.h5" contains the average rate of deformation at each pixel. Together, these datasets are about 17 GB in size, so it's inconvenient to download and work with them on our laptop. Thankfully, we can use CARC to make life easier!

```
In [1]: import h5py
import matplotlib.pyplot as plt
import numpy as np
from datetime import datetime
```



Viewing and int CARC

Large InSAR dataset on

The InSAR dataset "geo_...mErr_msk.h5" contains the complete time-series of deformation bet... the city of Jakarta, Indonesia. The file "geo_velocity_msk.h5" contains the average rate of deformation at each pixel. Together, these datasets are about 17 GB in size, so it's inconvenient to download and work with them on our laptop. Thankfully, we can use CARC to make life easier!

```
In [ ]: import h5py
import matplotlib.pyplot as plt
import numpy as np
from datetime import datetime
```

Trusted

Python [conda env:.conda-escape]

File Edit View Insert Cell Kernel Widgets Help



Run



Code



In [4]: *# first, read the map coordinates*



```
fname = data_path+'geo_timeseries_ERA5_ramp_demErr_msk.h5'
with h5py.File(fname,'r') as f:
    # read the metadata attributes:
    # to print all the attributes in the file (a lot!): print(f.attrs.keys())
    x0=float(f.attrs['X_FIRST'])
    dx=float(f.attrs['X_STEP'])
    nx=int(f.attrs['WIDTH'])
    y0=float(f.attrs['Y_FIRST'])
    dy=float(f.attrs['Y_STEP'])
    ny=int(f.attrs['LENGTH'])
    rlon=float(f.attrs['REF_LON'])
    rlat=float(f.attrs['REF_LAT'])
    # so, we have the x0 and dx, but need to turn them into vectors (x0,x1,
    lonvec = [x0 + i*dx for i in range(nx)]
    latvec = [y0 + i*dy for i in range(ny)]
```

[N] Means finished

Trusted

Python [conda env:.conda-escape]

File Edit View Insert Cell Kernel Widgets Help



Run



Code



In [*]: *# now, read some of the timeseries images and plot them in a grid (3x3)*



```
fname = data_path+'geo_timeseries_ERA5_ramp_demErr_msk.h5'
```

```
with h5py.File(fname, 'r') as f:
```

```
    # Read the dates from the file
```

```
    st = time.time()
```

```
    dates = [date.decode('utf-8') for date in f['date'][:]]
```

```
    timestep0 = f['timeseries'][0, :, :]
```

```
    print('Time to read dates:', round(time.time() - st, 3), 's')
```

```
    # Calculate nearly-evenly spaced timesteps
```

```
    st = time.time()
```

```
    total_timesteps = len(dates)
```

```
    spacing = total_timesteps // 9 # 9 is 3x3
```

```
    selected_t_indices = np.arange(0, total_timesteps, spacing)[:9] # Get
```

```
    print('Time to calculate nearly-evenly spaced timesteps:', round(time.t
```

* Means running

Hopper took 100 s
Wheeler 250 s.

```
/users/vanilla/workshops/escape/kml_functions.py:25: RuntimeWarning: Mean of empty slice
downsampled[i, j] = np.nanmean(block)
```

Time: 20151120



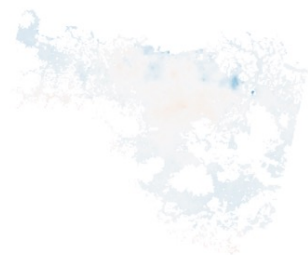
Time: 20170407



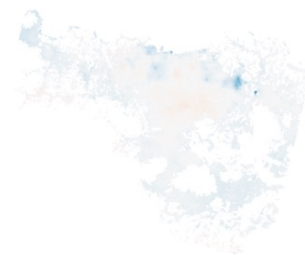
Time: 20171121



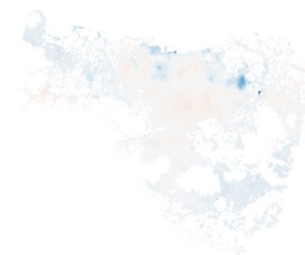
Time: 20180719



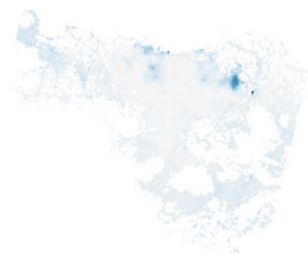
Time: 20190220



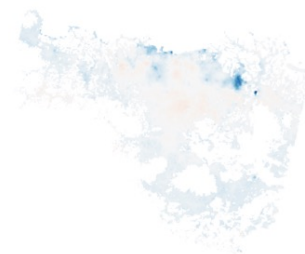
Time: 20191018



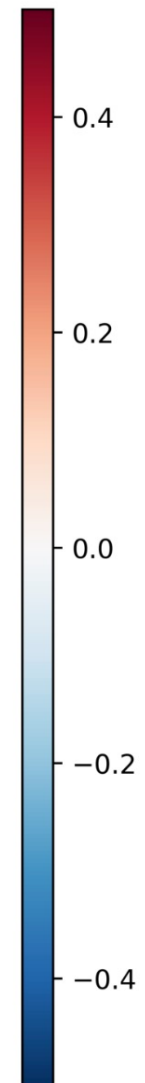
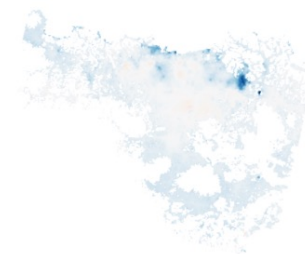
Time: 20200813



Time: 20210410



Time: 20220312



Shut down your jupyterhub server

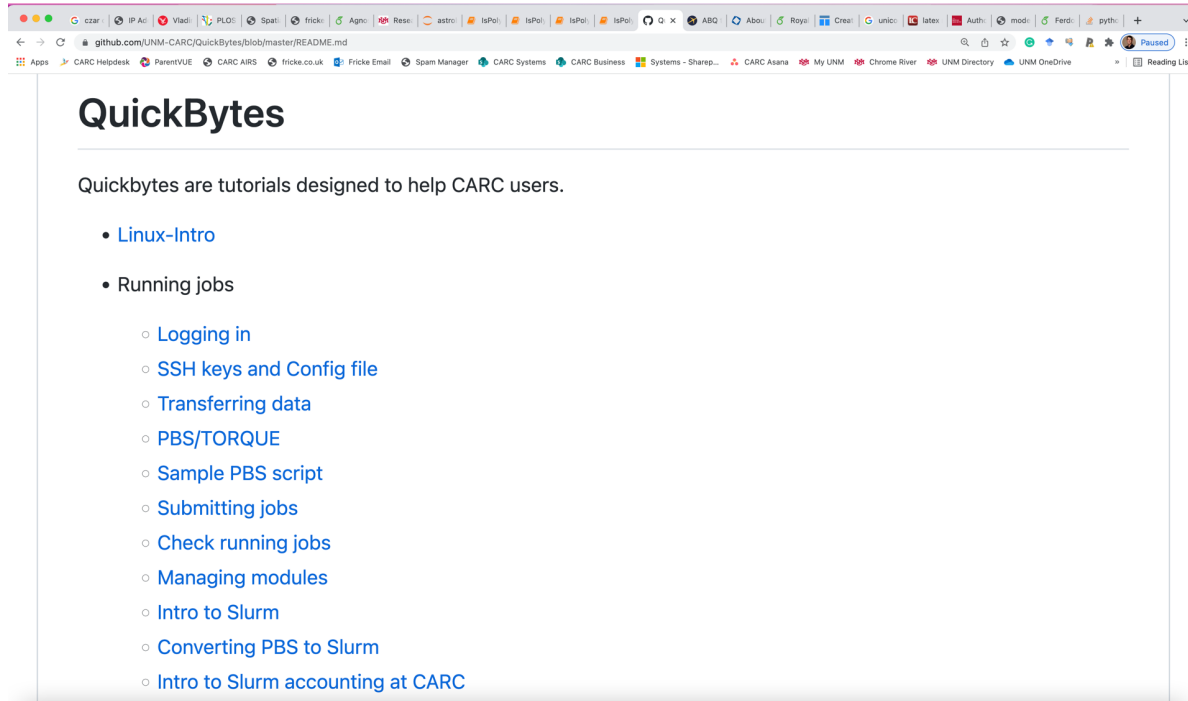
The image shows a JupyterHub interface. At the top, there's a header with the JupyterHub logo, the text "fit_fault_model", and "Last Checkpoint: an hour ago (unsaved changes)". To the right are "Logout" and "Control Panel" buttons. Below this is a menu bar with "File", "Edit", "View", "Insert", "Cell", "Kernel", "Widgets", and "Help". A "Trusted" badge and the environment "Python [conda env:.conda-fit_fault_model]" are also visible. The main area contains a notebook titled "Fault modeling example" with the subtitle "Made for ESCAPE program - 2022". The first code cell is active, showing the following code:

```
In [1]: 1 # import statements - general statements
        2 import numpy as np
        3 import matplotlib.pyplot as plt
```

At the bottom, there's a navigation bar with the JupyterHub logo and links for "Home", "Token", and "Admin". Below this bar are two buttons: a red "Stop My Server" button and a blue "My Server" button. A large blue arrow points from the "Control Panel" button in the top right to the "Stop My Server" button at the bottom.

CARC Resources

- Tutorial Videos
- Written Tutorials

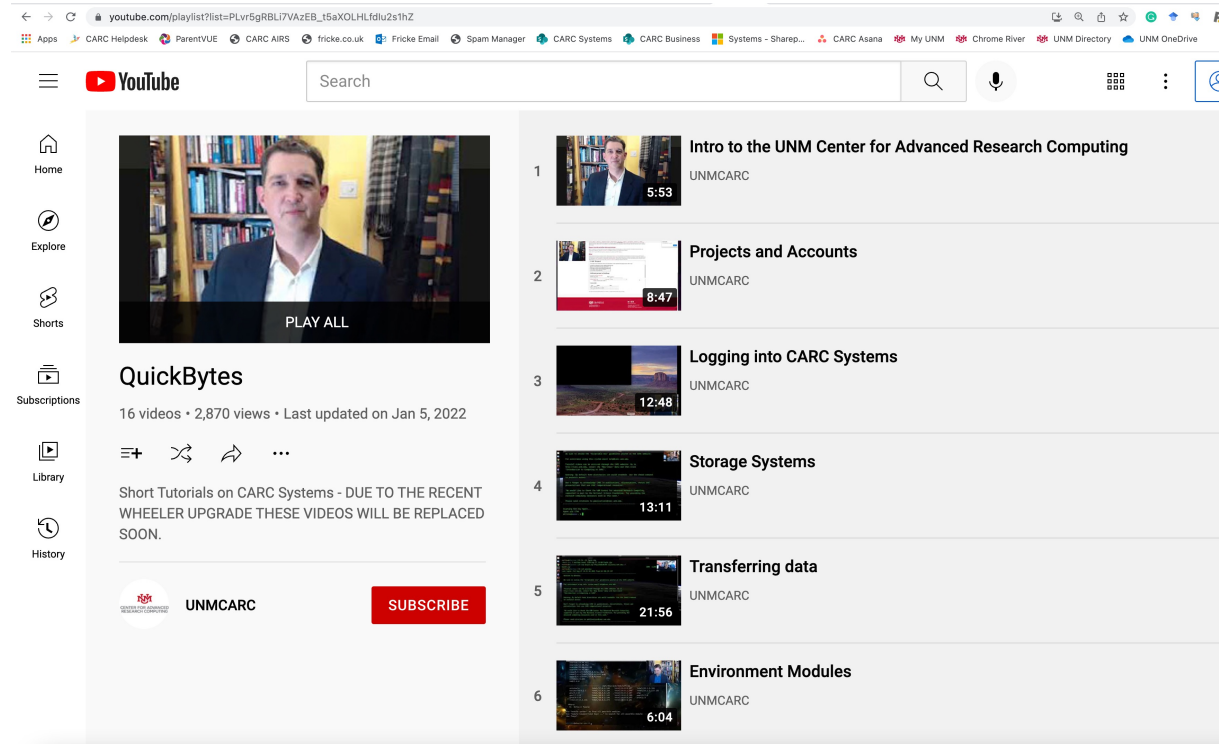


The screenshot shows a web browser displaying the GitHub repository for UNMC-CARC/QuickBytes. The repository name is visible in the address bar. The page title is "QuickBytes". Below the title, a description states: "Quickbytes are tutorials designed to help CARC users." A list of tutorial topics is provided, including "Linux-Intro" and "Running jobs". The "Running jobs" section is expanded, showing a list of sub-topics: "Logging in", "SSH keys and Config file", "Transferring data", "PBS/TORQUE", "Sample PBS script", "Submitting jobs", "Check running jobs", "Managing modules", "Intro to Slurm", "Converting PBS to Slurm", and "Intro to Slurm accounting at CARC".

QuickBytes

Quickbytes are tutorials designed to help CARC users.

- [Linux-Intro](#)
- [Running jobs](#)
 - [Logging in](#)
 - [SSH keys and Config file](#)
 - [Transferring data](#)
 - [PBS/TORQUE](#)
 - [Sample PBS script](#)
 - [Submitting jobs](#)
 - [Check running jobs](#)
 - [Managing modules](#)
 - [Intro to Slurm](#)
 - [Converting PBS to Slurm](#)
 - [Intro to Slurm accounting at CARC](#)



The screenshot shows a YouTube playlist page for UNMC-CARC. The playlist is titled "QuickBytes" and contains 16 videos with 2,870 views, last updated on Jan 5, 2022. The video thumbnails and titles are listed on the right side of the page. The UNMC-CARC channel logo and a "SUBSCRIBE" button are visible below the video player.

QuickBytes

16 videos • 2,870 views • Last updated on Jan 5, 2022

- 1 [Intro to the UNM Center for Advanced Research Computing](#) UNMCARC 5:53
- 2 [Projects and Accounts](#) UNMCARC 8:47
- 3 [Logging into CARC Systems](#) UNMCARC 12:48
- 4 [Storage Systems](#) UNMCARC 13:11
- 5 [Transferring data](#) UNMCARC 21:56
- 6 [Environment Modules](#) UNMCARC 6:04

Getting Help

help@carc.unm.edu

Office hours